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Implementing AI-Powered Content Management Pipelines for Infolep and InfoNTD Knowledge Platforms

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Statement of Originality

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Abstract

The accelerating growth of biomedical literature presents significant challenges for manual curation on specialized knowledge platforms such as Infolep and InfoNTD, which focus on leprosy and neglected tropical diseases (NTDs). This thesis explores the application of artificial intelligence methods to automate and scale content management for these platforms. Two central classification tasks were addressed: identifying whether a publication is relevant to NTDs, and if so, assigning it to one or more specific disease categories. For those tasks, expert-annotated data from Infolep/InfoNTD was combined with large-scale records from PubMed. Three AI approaches - support vector machines (SVM), fine-tuned BERT, and zero-shot Gemini 2.0 Flash - were used and assessed both individually and in ensembles. Results showed that while BERT achieved the highest classification performance, SVMs offered comparable outcomes with significantly greater computational efficiency, making them well-suited for practical deployment. The Gemini LLM offered adaptability and strong out-of-the-box performance, but did not consistently reach the performance of other tested methods. A proof-of-concept tool was presented to automate literature retrieval, classification and export for human review. Overall, this work demonstrates that AI-powered pipelines can improve the scalability and comprehensiveness of literature curation workflows for NTD knowledge platforms.

Contents

1 Introduction

The volume of biomedical literature being published has been increasing rapidly, posing a growing challenge for researchers and health professionals who must remain up to date with the latest developments (Jin et al., 2024). The PubMed database alone contains over 36 million records, with more than 1.5 million new entries added annually. Traditional search engines are often insufficient for conducting thorough, domain-specific queries (Novoa et al., 2023). In response, specialised knowledge platforms such as DisGeNET and LitCovid have been developed to support curated access to focused biomedical subdomains (Jin et al., 2024; Chen et al., 2021).

Infolep and InfoNTD are two such platforms, designed to support access to knowledge on leprosy and neglected tropical diseases (NTDs), respectively. Jointly referred to in this thesis as the Info Platforms, they are maintained by the international NGO No Leprosy Remains (NLR). Their mission is to strengthen the capacity of researchers, program managers, policymakers, and healthcare professionals by providing curated access to relevant literature and resources (Infolep, n.d.; InfoNTD, n.d.). Currently, the content for the platforms is manually curated by experts who identify relevant publications from various online sources and classify them using tags such as topic areas, diseases and geographic focus. However, as the volume of literature continues to grow, this expert-driven curation model has become increasingly resource-intensive and difficult to sustain. Financial and human resource limitations create an urgent need for more efficient content management pipelines that reduce reliance on manual labour while maintaining high-quality curation.

Recent advances in artificial intelligence offer promising opportunities to automate aspects of this process. Traditional machine learning methods, such as support vector machines (Aphinyanaphongs et al., 2005; Chen et al., 2021), transformer-based models like BERT (Devlin et al., 2018; Lee et al., 2019; Gu et al., 2022), and more recently, large language models (LLMs) such as GPT-4 (Guo et al., 2024; Chen et al., 2025), have demonstrated strong performance in various biomedical text classification tasks. Hybrid and ensemble methods that combine these models have shown improved robustness and generalisability in literature screening and topic classification (Sakai & Lam, 2025b; Chen et al., 2021). While these approaches have yet to be systematically applied in the context of NTD literature, they provide a solid foundation for designing scalable solutions.

The aim of this thesis is to explore how these AI-based methods can be adapted and applied to support the content management needs of the Infolep and InfoNTD knowledge platforms. Specifically, whether AI pipelines can assist in identifying relevant publications and tagging them with appropriate disease labels, to reduce the manual effort required to maintain the platforms and to accelerate the identification of relevant literature. The remainder of the thesis is structured as follows. Chapter 2 presents a review of the relevant literature, including traditional and modern approaches to text classification. Chapter 3 outlines the specific contributions to the literature. Next, Chapter 4 details the methodology, covering the data sources, dataset construction process, and experimental setup. In Chapter 5, the results are presented. The limitations and further research are discussed in Chapter 6. Finally, Chapter 7 describes the deliverable proof-of-concept tool developed for the NLR and Chapter 8 concludes the findings.

2 Literature review

The volume of biomedical and clinical research literature published is growing at a fast rate, making manual curation increasingly impractical (Chen et al., 2021; Jin et al., 2024; Chen et al., 2025). Every month, 130 thousand articles are added to the PubMed library alone. With thousands of potential publication topics in each domain, human annotation is difficult and expensive, creating a large curation bottleneck (Sadat & Caragea, 2022). The challenge of selecting papers and annotating them is not unique to Infolep and InfoNTD knowledge platforms. LitCovid is a knowledge platform developed during the COVID-19 pandemic in 2020 (Chen et al., 2020), which, as of May 2025, has identified 456 thousand papers relevant to the virus and classified them into eight main topics (LitCovid, n.d.). The maintainers of this hub also employed a mix of AI techniques to manage this effort, as manual curation alone could not keep up with the influx of 10 thousand new related articles being published each month during the pandemic (Chen et al., 2021).

The growing interest in artificial intelligence has increasingly extended into healthcare text classification, where it is helping to automate, scale and enhance the accuracy of traditionally labour-intensive expert processes. Researchers have employed a range of traditional machine learning methods, such as support vector machines, as well as modern neural approaches, including transformer-based models like BERT. More recently, large language models (LLMs) accessible through APIs, such as GPT-4, have enabled fast annotation across diverse corpora, even in data-scarce scenarios (Sakai & Lam, 2025a). The following sections will review these methods in more detail, covering traditional techniques, transformer-based models, LLM APIs, ensemble approaches and metadata extraction strategies.

2.1 Traditional approaches

Traditional methods for text classification include approaches like rule-based systems (Apté et al., 1994), support vector machines (SVMs) (Joachims, 1998), decision trees (Lewis and Ringuette, 1994), and ensemble methods such as random forests (Xu et al., 2012). While relatively simple, they have been found useful in the efforts to manage the growing volume of literature.

For example, Aphinyanaphongs et al. (2005) used polynomial SVMs to classify medical article abstracts into clinically relevant categories such as treatment, diagnosis, etiology and prognosis. Compared to PubMed search filters, this approach doubled the retrieval precision e.g. from 9% to 18% in the treatment category. Similarly, Chen et al. (2021) used SVMs with bag-of-words features in the LitCovid platform to determine the relevance of articles to COVID-19. Du et al. (2024) used XGBoost to automate abstract screening in systematic literature reviews in two medical domains, reporting good results. The authors highlight the fast training times and low computational overhead as positives of using these approaches when they can achieve competitive results. However, they have several notable limitations. Such methods rely on engineered features and do not inherently capture the contextual nuances in biomedical text, like acronyms, negations and jargon (Sakai & Lam, 2025a). As those texts are characterised by niche words, the performance also suffers in the absence of specialised word embeddings for biomedical vocabulary (Chen et al., 2019). Maintaining the required feature spaces becomes increasingly challenging as the literature grows, and handling a large number of possible classification labels can be computationally demanding. Despite these drawbacks, traditional models still serve a valuable role in newer literature as performance benchmarks in comparative evaluations (Sakai & Lam, 2025b; Raja et al., 2024).

2.2 Transformer-based methods

Transformer-based language models have significantly advanced natural language processing by introducing an attention mechanism that considers the relevance of different words in a sequence when producing embeddings, allowing for more nuanced understanding of text (Vaswani et al., 2017). A prime example used in the relevant literature is BERT (Bidirectional Encoder Representations from Transformers), which is a transformer encoder with 12 layers and 110 million parameters in its base version (Devlin et al., 2018). BERT's manageable size makes fine-tuning the model parameters on domain-specific corpora accessible, allowing for improved outcomes in specific use-cases. Examples of the model's tailored versions include BioBERT, fine-tuned on various biomedical texts (Lee et al., 2019) and PubMedBERT, trained on PubMed data (Gu et al., 2022). BERT models remain widely used in literature classification not only because of performance improvements, but also due to practical advantages. Many model variants are open-source and freely available. They are also lightweight compared to other LLMs, allowing local deployment with standard infrastructure (Sakai & Lam, 2025a). Such models also support multi-label outputs, eliminating the need for training multiple classifiers, as required by traditional approaches.

Recent literature illustrates the versatility of these models across classification tasks. Chen et al. (2022) developed LitMC-BERT, a fine-tuned BERT variant for classifying COVID-19 articles into eight categories for the LitCovid platform. Their model replaced the previously used eight binary classifiers with a single multi-label setup using a sigmoid activation layer. This increased F1 scores by ~5% and reduced inference time fivefold, allowing for faster publication processing. Rei et al. (2024) built a similar classifier to determine if a scientific or news article addresses a rare or non-rare disease, focusing on neurodevelopmental disorders. In their approach, the authors used the RoBERTa encoder (Liu et al., 2019) to tokenise the text input into subwords, which is particularly useful for improving the model performance on rare vocabulary. They also enhanced the model input with the MeSH terms retrieved from their data source, despite acknowledging known gaps in the coverage for rare diseases. This approach yielded strong results, especially on scientific article abstracts. Raja et al. (2024) focused on classifying ophthalmology papers into 19 topic groups and comparing various BERT variants. They found a fine-tuned BioBERT to perform best. Another interesting observation of the authors was that applying the classification solely on the publication titles closely approximated the performance obtained from also using abstracts - highlighting potential efficiency gains. They also decided against using large language models like Llama 2, due to their computational requirements, underscoring the practical need for resource-efficient methods. Sadat and Caragea (2022) took on the challenge of classifying articles into a large set of 1,233 hierarchical topics using fine-tuned BERT and SciBERT models. While their results showed that transformer-based models consistently outperformed traditional neural methods, the overall performance was still modest, with the best macro F1 scores reaching only around 34%. This highlighted the difficulty of applying these methods to highly granular classification schemes with numerous labels, despite their relative advantage over previous approaches. Overall, transformer-based methods offer strong performance, adaptability and operational feasibility, which made them a popular choice in biomedical NLP research. Nonetheless, they still require labelled training data and struggle with choosing across a large set of labels.

2.3 Large Language Models

The latest advance in medical text classification is the use of Large Language Models such as GPT-4, which are extremely large transformer models (OpenAI, 2023). These models, trained on massive text corpora, can be prompted to perform classification without dedicated task training, essentially asking the LLM to directly infer the categories of a given text by providing a task description or examples in the prompt (Sakai & Lam, 2025a). Compared to previous approaches, such models are often too computationally demanding to be run locally, hence are applied through paid API services. The consistency of their output is also not guaranteed, so additional post-processing steps of their responses may be needed. Recent studies demonstrated strong results from this approach.

Guo et al. (2024) explored the use of GPT-4 to screen papers for clinical review using a zero-shot approach (not providing examples in the prompt), reporting high performance on an expert-annotated dataset. Notably, the authors included an interpretability step, asking the model to justify its decision. The LLM often corrected its misclassifications when answering, suggesting that multi-stage verification could improve reliability. Sanghera et al. (2024) conducted a broad comparison of LLMs such as GPT-4o, Llama 3 70B, Gemini 1.5 Pro, and Claude Sonnet 3.5 for abstract screening. In their experiment, LLM ensembles outperformed both individual models and human reviewers in sensitivity, precision and balanced accuracy. The authors emphasised the importance of prompt engineering, which had a significant impact on the outcomes. Delgado-Chaves et al. (2025) further examined literature screening by evaluating 18 LLMs against human selections. While GPT-4o delivered the best results overall, smaller models such as Gemma2-9B performed comparably to larger ones, indicating that size does not necessarily correlate with effectiveness. This is particularly important for situations with limited resources. Finally, Chen et al. (2025) benchmarked different models across 18 medical datasets. Their suggestions for text classification include fine-tuning BERT models when labelled training data is plentiful. For data-scarce situations, such as broad topic classification, the authors recommend using GPT-4 with dynamic few-shot prompting. Overall, LLMs present a powerful new direction for biomedical literature classification, improving workflows through self-verification and not requiring dedicated task training. However, their practical deployment is constrained by their consistency and cost.

2.4 Hybrid approaches

Recent work has shown that hybrid approaches combining multiple AI systems can help balance the advantages and disadvantages of each approach. Sakai and Lam (2025b) built an ensemble method called QUAD-LLM-MLTC for the task of multi-label topic classification in healthcare. Their framework leverages the complementary strengths of four models in a pipeline: a BERT-based model first extracts key tokens from the text, then a PEGASUS transformer generates an augmented summary. Next, GPT-4o creates three predictions, based on the original text, the extracted key tokens, and the augmented text, which are used for hard voting. In parallel, a BART model is given the original text and estimates probabilities for each topic classification. Finally, a meta-classifier using Linear Support Vector Machines determines the final topic prediction. In the experiments, QUAD-LLM-MLTC delivered strong results, improving on the performance of a standalone zero-shot GPT-4o. The performance was also more consistent across topics. Another hybrid approach was used by Chen et al. (2021) for the manual curation of papers for the LitCovid database. The predictions of the previously mentioned SVMs and LitMC-BERT, together with other approaches such as deep learning models based on word embeddings, were averaged and used as a reference during human review through a user-friendly curation system. These examples illustrate how hybrid systems can lead to improved accuracy and robustness, making them a promising direction for more complex biomedical text classification tasks.

2.5 Metadata extraction

In addition to classifying publication abstracts to determine if they are on the topic of neglected tropical diseases, AI methods could also be used to assist in the side task of metadata extraction for the InfoNTD and Infolep platforms. One common approach is to treat metadata tagging as a sequence labelling problem. Sammet and Krestel (2023) researched using BERT to automatically extract keywords from article titles and abstracts in a domain-specific library. They fine-tuned the model to perform token-level marking if a word is part of a keyphrase, identifying important technical terms, which should be added to a controlled vocabulary list. This method outperformed other extraction baselines, showing that BERT's context understanding helps determine relevant terms, even without exact string matches. Another emerging powerful technique is LLM driven metadata generation. Such models can not only extract words from the text, but also propose other keywords capturing the document's essence. Umair et al. (2024) reviewed recent advances in using pre-trained models for this task and concluded that modern LLMs such as GPT-4 and T5 can perform well on generating such phrases, for example, inferring "neglected tropical diseases" as a keyword for a paper that discusses multiple NTDs without using that exact phrase. According to the authors, transformer models have been setting new benchmark records on keyphrase tasks, making them prime candidates for automation use cases.

In summary, biomedical literature classification has progressed from traditional machine learning methods to transformer-based models and, more recently, Large Language Models (LLMs). While earlier approaches offer simplicity, transformer models like BERT provide stronger contextual understanding. LLMs enable flexible, zero-shot classification, though they come with cost and consistency challenges. Hybrid systems and metadata extraction techniques further enhance performance and scalability, offering promising solutions for automating literature curation in complex domains.

3 Contributions

The application of artificial intelligence to biomedical literature curation is a growing area of research, driven by the increasing need to manage ever-expanding scientific repositories. This thesis contributes to that effort by developing and evaluating AI-powered content management pipelines tailored to the needs of the Info Platforms. Hence, the central research question guiding this work is: **To what extent can AI models support scalable, accurate, and practically deployable publication classification for Infolep and InfoNTD platforms?**

To guide this research, two main classification tasks are defined:

(**1**) **Inclusion classification** - identifying whether a publication is relevant to the scope of the neglected tropical diseases.

(**2**) **Disease classification** - determining which of the 19 monitored neglected tropical diseases the publication is about.

For both tasks, the following sub-questions are explored:

1. How can a robust dataset be constructed for classifying NTD-related literature?
2. How do different model families - traditional (SVM), transformer-based (BERT) and large language models (Gemini 2.0 Flash) - perform on the NTD literature classification tasks, and can their ensemble improve robustness?
3. What common error patterns can be observed across the tested models?
4. How can the pipelines be integrated into a practical, automated workflow to support ongoing curation of new publications?

To address these questions, a complete content classification framework was developed and evaluated, spanning from dataset construction to practical deployment. The dataset construction process builds on prior works such as LitCovid (Chen et al., 2021) and Rei et al. (2024). Expert-labelled records were combined with large-scale biomedical text corpora, resulting in four experimental datasets supporting both binary and multi-label classification tasks. While other studies have addressed specific domains or diseases, no prior work has developed large-scale datasets focused on neglected tropical diseases. Three models were implemented, reflecting key directions in the biomedical natural language processing literature. Support vector machines and fine-tuned BERT models were adapted to the defined NTD classification tasks. Gemini 2.0 Flash - a fast large language model accessible through an API - was tested in a zero-shot configuration. While recent studies have explored the use of LLMs (Sanghera et al., 2024; Guo et al., 2024), smaller-sized LLMs and the Gemini family of models have not previously been tested in biomedical literature classification tasks. Inspired by Sakai & Lam (2025b), an ensemble strategy was also implemented to test whether model combinations improve robustness. In addition to performance evaluation, a targeted error analysis was conducted to examine failure patterns across the tested models. Finally, a deployable curation tool was developed as a proof-of-concept for the NLR team, considering the uncovered trade-offs between performance and practicality.

4 Methodology

4.1 Data sources

Biomedical text classification requires not only large quantities of textual data but also reliable annotations to identify relevant categories. While information on many scientific publications is publicly accessible through online repositories, expert-labelled datasets for domain-specific tasks are rare. Consequently, most of the relevant literature relies on bespoke datasets manually labelled by experts for the needs of a particular study, rather than using a standardised benchmark. The LitCovid dataset, used by Chen et al. (2021), was compiled through expert review of COVID-19-related publications from PubMed. Raja et al. (2024) created the RenD dataset by manually labelling 1,000 PubMed articles related to retinal diseases. Sakai and Lam (2025b) used the Hallmarks of Cancer dataset developed by Baker et al. (2016), which was made by annotating 1,499 PubMed abstracts. Another approach was taken by Rei et al. (2024), who labelled PubMed abstracts based on their assigned MeSH terms. This allowed the authors to create a large, low-cost corpus, however, it also increased the risk of mislabelling because of the gaps in the available MeSH terms.

Inspired by these approaches, this thesis adopts a hybrid data strategy to support two classification tasks: identifying whether a paper is related to NTDs and, if so, determining which specific diseases it addresses. They are supported by two primary data sources. Firstly, expert-annotated records were obtained from the backend of the Infolep and InfoNTD knowledge platforms. Secondly, this data was supplemented with a large corpus of biomedical publications extracted from PubMed with their related Medical Subject Headings (MeSH) terms. Additionally, a set of API connections to PubMed, Crossref and ArXiv repositories was developed to enable automated retrieval of newly published papers, ensuring practical applicability of the proposed workflows. Each of these components is described in detail in the following subsections.

4.1.1 Infolep/InfoNTD backend data export

To make this research possible, the NLR provided access to a data export of the backend running the Infolep and InfoNTD knowledge platforms, as of April 2025. This export was then converted to an SQLite database to facilitate content exploration. Relevant tables were identified and joined to construct a unified data source containing **39,070** unique scientific publications from both platforms. Each entry includes metadata such as publication type, title, abstract, year, secondary title, related URL, Digital Object Identifier (DOI), language, knowledge portal (Infolep, InfoNTD or both), topic area, related geographical regions, diseases, and keywords. While some records are missing specific fields, such as abstracts or DOI, every entry contains a title and at least one disease label. Overall, 23,737 (61%) records include abstracts that could be used for model training. The availability of abstracts is not uniformly distributed across time: publications from earlier decades are much more likely to be missing an abstract, whereas more recent entries tend to include one (Figure 1).

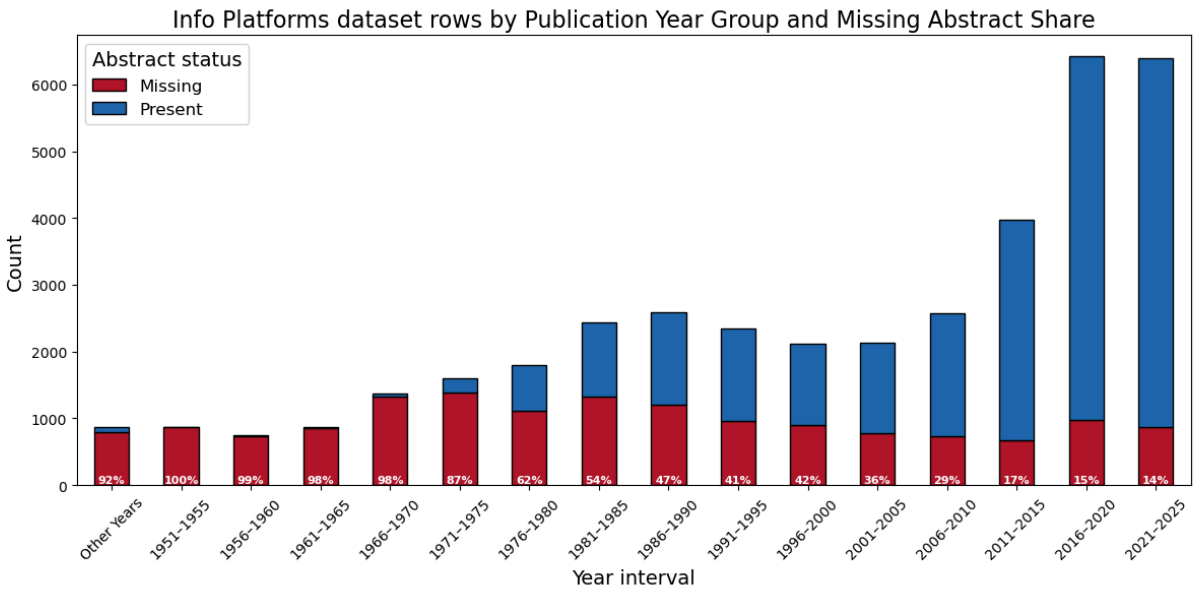


Figure 1: Share of entries with missing abstracts in the Info Platforms dataset over time.

The data covers 19 neglected tropical diseases, though the disease distribution is heavily skewed. Infolep, which focuses exclusively on leprosy, has operated for a longer period than InfoNTD and contributes the majority of older publications in the dataset. As a result, leprosy is vastly overrepresented relative to other diseases. In contrast, InfoNTD includes a broader set of diseases and contains a higher proportion of recent publications, which can be seen in Figure 2.

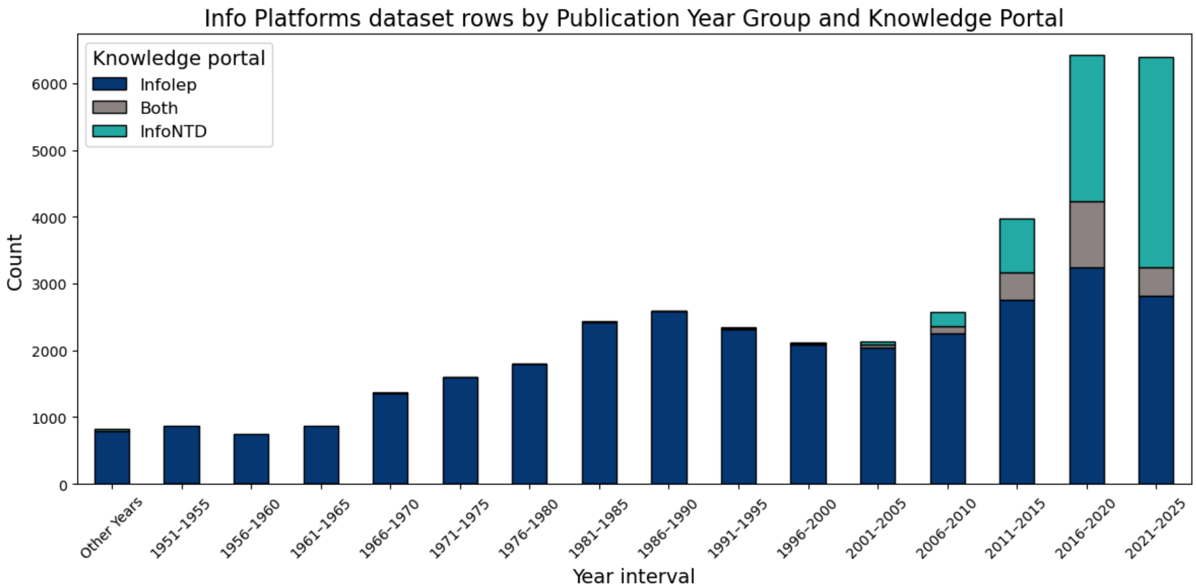


Figure 2: Distribution of the publications in the Info Platforms data source in time and by platform.

4.1.2 PubMed bulk data download

PubMed is a publicly accessible resource maintained by the U.S. National Library of Medicine, containing information on biomedical and life sciences literature (PubMed, 2025b). It is widely used for academic and clinical research due to its broad domain coverage. For this research, the PubMed 2025 Annual Baseline dataset was used, which is a full snapshot of the database as of January 2025 in the XML format (PubMed, 2025a). This dataset was processed into a structured format suitable for text classification, resulting in a corpus of over 36.5 million records. Of these, 25.3 million (69%) include publication abstracts. As with the Info Platforms dataset, the availability of abstracts is strongly dependent on the publication date - older publications are significantly less likely to include abstracts, while coverage improves for newer entries (Figure 3).

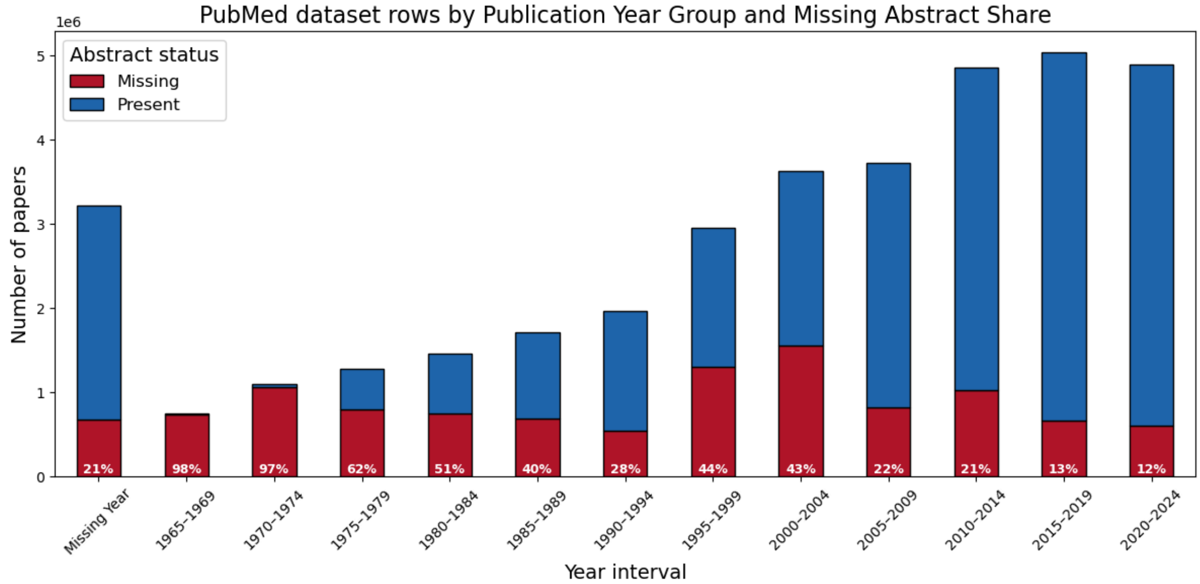


Figure 3: Share of entries with missing abstracts in the PubMed 2025 Annual Baseline dataset over time.

The columns in the processed dataset include the title, abstract, authors, language, publication year and month, publication type, affiliated agency, MeSH terms and DOI. Of particular interest for this project are the MeSH terms, which serve as additional curated disease annotations. However, their coverage is incomplete, with 5.4 million (15%) entries missing these annotations. Notably, 3.2 million of those correspond to publications for which the publication year is also missing. Among the remaining 2.2 million entries, most were published from the year 2000 onwards (Appendix A.1). Furthermore, 29,267 records from the Info Platforms dataset were matched within the PubMed data based on the title and DOI, underscoring PubMed's strong coverage of publications relevant to the neglected tropical disease domain.

4.1.3 API connections

For the practical use of the developed workflow, API connections to three major literature repositories were established: PubMed, Crossref, and arXiv. These connections allow for regular retrieval of the latest scientific publications and can be used to continuously update and classify new papers in an automated fashion.

The PubMed data is downloaded through their Daily Update uploads (PubMed, 2025a). On average, around 130 thousand new entries are added to this repository each month. Crossref is a non-profit organisation that maintains a registry of scholarly content metadata through its network of member publishers (Crossref, 2021). Through its REST API, users can access metadata for a wide range of academic content across disciplines, including journal articles, conference proceedings and reports. Every month, the repository adds around 418 thousand new entries. While there is a considerable overlap with PubMed for biomedical articles, Crossref records are less curated and do not contain MeSH annotations. However, they may include entries related to NTDs which are missing from PubMed. Lastly, arXiv is a research preprint sharing platform primarily focused on physics, mathematics, computer science and quantitative biology (arXiv, n.d.). It provides free access to early-stage research before formal peer review, contributing roughly 22 thousand new entries monthly. While its domain coverage is narrower, it expands the searched sources with the most recent research. Together, these three sources ensure high coverage and diversity in the retrieved literature for the proof-of-concept workflow developed.

4.2 Experimental dataset construction

To support the classification tasks defined in this thesis, four experimental datasets were constructed from the previously described sources. These datasets serve both as training material for the machine learning models and a basis for evaluating their performance, aiding the final workflow design decisions. The construction process was guided by the two key objectives of the study: (1) identifying whether a publication is relevant to NTDs, and (2) determining which specific NTDs are addressed in each selected publication. Each classification task was supported by a dedicated dataset pair, designed to reflect both balanced experimental conditions and more realistic deployment scenarios.

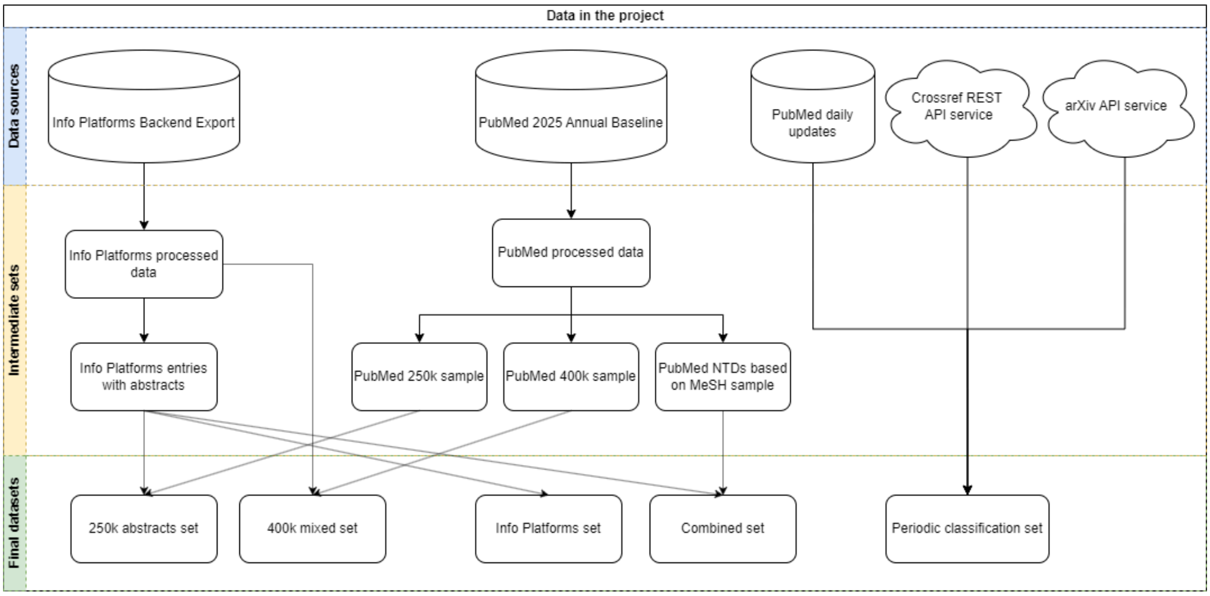


Figure 4: Dataset construction pipeline used in the thesis.

4.2.1 Inclusion classification task

The objective of the inclusion classification task was to determine whether a given scientific publication is relevant to the scope of the Info Platforms. Formulated as a binary text classification problem, this task assigns a target label of 1 to relevant entries and 0 to non-relevant ones. Two datasets were constructed for this purpose, both using the Info Platforms export as the positive-class foundation. The first dataset includes only entries with available abstracts, while the second retains all entries regardless of abstract availability, more accurately reflecting real-world deployment scenarios. Given the scale of the PubMed corpus, it was not feasible to use it in its entirety. Instead, two negative-class samples were drawn, sized at approximately 250 and 400 thousand entries, respectively. These correspond to an approximate 1:10 relative ratio to the Info Platforms dataset. To prioritise more recent publications, sampling was conducted using an exponentially decaying probability distribution over publication years. Specifically, for a given year y, the sampling weight w_y was defined as:

$$w_y = \frac{e^{\alpha(y-y_0)}}{\sum_{t=y_0}^{y_n} e^{\alpha(t-y_0)}}$$

Where y_0 is the starting year (1965 for PubMed), y_n is the final year (2024), and α is the exponential growth parameter (set to 0.07). This formulation ensures that more recent years are sampled with higher probability, reflecting the practical need to classify newly published literature.

	250k abstracts set		400k mixed set	
	Train	Test	Train	Test
Positive Target	20,726	5,182	34,552	8,631
Negative Target	199,517	49,879	319,175	79,794

Table 1: Entry counts for the dataset splits used in the inclusion classification task.

To ensure the integrity of the negative class, all PubMed entries that matched the Info Platforms dataset based on either title or DOI were excluded from the sample. Initially, all PubMed records in the sample were assigned a target label of 0. However, to correct for mislabelled instances, entries containing any of the manually curated list of relevant MeSH terms (Appendix A.2) were reclassified as positive, ensuring alignment with NTD scope of the Info Platforms. For both datasets, an 80/20 train/test split was performed with stratification based on the target label to preserve class balance across splits. When needed by specific model architectures, a validation set was derived by reserving 10% of the training data. The test sets were constructed to be fully independent not only from their corresponding training set, but also the other training set. This ensures that the test set from the larger dataset can also serve as a reliable evaluation benchmark for models trained on the smaller dataset, without data leakage. Final counts for the dataset splits in the inclusion classification task are presented in Table 1.

4.2.2 Disease classification task

The second experimental task focuses on identifying which NTDs are addressed within a given publication. This is formulated as a multi-label classification problem, where each of the 19 diseases represented in the Info Platforms is treated as a separate binary label. Publications may be associated with one or more diseases simultaneously, reflecting the potential for overlap in disease relevance. The task assumes that all input publications are already within the scope of the Info Platforms (i.e. are relevant to NTDs). Given that disease-specific classification typically requires richer textual context, the dataset was limited to entries that include an abstract.

Two datasets were prepared for this task. The first one consists exclusively of the Info Platforms entries, which contain expert-curated disease labels. However, as shown in Table 2, this data is highly imbalanced: while some diseases, such as leprosy, have thousands of entries, others have only a few dozen. To partially mitigate this issue and improve model learning for underrepresented classes, a second dataset was constructed by augmenting the Info Platforms data with additional entries from the full PubMed corpus. Specifically, publications with MeSH terms matching the 19 target diseases were identified and added (Appendix A.2). This augmentation substantially increased the number of examples for several diseases - for instance, Zika virus increased from 95 to 6,109 entries - though for others the impact was more modest (e.g. Buruli Ulcer grew from 651 to 1,124 entries). While the additional data improves class support, it also introduces additional noise, since MeSH annotations are not as precise or reliable as the expert labels used in the original dataset. Therefore, performance differences between the two datasets were evaluated to assess the trade-off between quantity and label quality.

Both datasets were split into training and test sets using an 80/20 split, with stratification performed across all disease labels to maintain representative class distributions. A validation set of 10% of the training data was also used, where required. As with the inclusion classification task, care was taken to ensure that the test sets are mutually exclusive from both training sets, allowing them to serve as common evaluation benchmarks across different experimental configurations. Final label distributions and entry counts are reported in Table 2. The three rows with the highest impact of adding MeSH terms are highlighted in green and three with the lowest in red.

Disease	Info Platforms set		Combined set	
	Train	Test	Train	Test
Leprosy	13,995	3,495	14,226	3,557
Lymphatic filariasis (Elephantiasis)	935	226	1,184	286
Schistosomiasis	866	226	6,116	1,585
Soil-transmitted helminths	754	192	2,390	612
Leishmaniasis	676	169	15,569	3,887
Onchocerciasis	539	119	2,354	576
Buruli Ulcer	521	130	911	213
Trachoma	515	122	1,667	407
Chagas disease	373	88	7,855	1,987
Snakebite envenoming	280	69	4,999	1,206
Human African trypanosomiasis (Sleeping sickness)	222	53	3,674	939
Yaws	206	47	412	91
Podoconiosis	192	46	192	46
Mycetoma	176	43	1,101	241
Dracunculiasis (guinea-worm disease)	149	31	423	100
Scabies	130	36	1,562	421
Zika virus	76	19	4,886	1,223
Chromoblastomycosis	50	12	487	105
Noma	29	7	243	60
Total rows	**17,320**	**4,331**	**66,289**	**16,574**

Table 2: Entry counts for the dataset splits used in the disease classification task. The rows with the highest change in the Combined set are highlighted.

4.3 Pipelines

Three distinct modelling approaches were selected to construct experimental pipelines for both classification tasks addressed in this thesis: determining entry inclusion and identifying specific diseases. These approaches align with recent developments in biomedical text classification literature. They comprise support vector machines (SVMs) as a baseline (Chen et al., 2021; Sakai & Lam, 2025b), fine-tuned BERT as a transformer-based model (Chen et al., 2022; Rei et al., 2024; Raja et al., 2024; Sadat & Caragea, 2022), and Gemini 2.0 Flash as an API large language model (Guo et al., 2024; Sanghera et al., 2024; Chen et al., 2025). Collectively, these methods were selected to provide a comprehensive coverage of the prominent model architectures in use. Additionally, their potential as components of an ensemble was investigated, inspired by successful hybrid approaches demonstrated in prior research (Sakai & Lam, 2025b; Chen et al., 2021).

4.3.1 Support Vector Machines Pipelines

SVMs are supervised machine learning models widely used for classification tasks, which work by finding an optimal hyperplane that maximises the margin separating different classes (Joachims, 1998). Due to their minimal resource requirements and ease of deployment, SVMs are employed in this research as a baseline approach. Their lightweight nature makes them particularly suitable for handling large text corpora, such as those used in this thesis. Despite their strengths, SVMs do not inherently capture contextual information, potentially limiting their effectiveness on complex texts.

The components of the SVM pipelines were implemented in Python using the scikit-learn library (Pedregosa et al., 2011). The features for the SVMs were generated using term frequency-inverse document frequency (TF-IDF) vectorisation, which quantifies the importance of words in entries relative to their prevalence across the entire dataset

(Robertson, 2004). In this step, English stop words were removed to reduce noise and focus on meaningful terms. The models were trained using a linear support vector classifier (LinearSVC), with optimal hyperparameter values for the regularisation parameter c selected through GridSearch cross-validation. After identifying the best-performing hyperparameter, predictions for the final model were calibrated using CalibratedClassifierCV to provide probabilistic outputs, enhancing interpretability. For the multi-label disease classification task, a OneVsRestClassifier strategy was implemented to train independent binary classifiers for each disease label. The entire SVM pipeline for both classification tasks is shown in Figure 5.

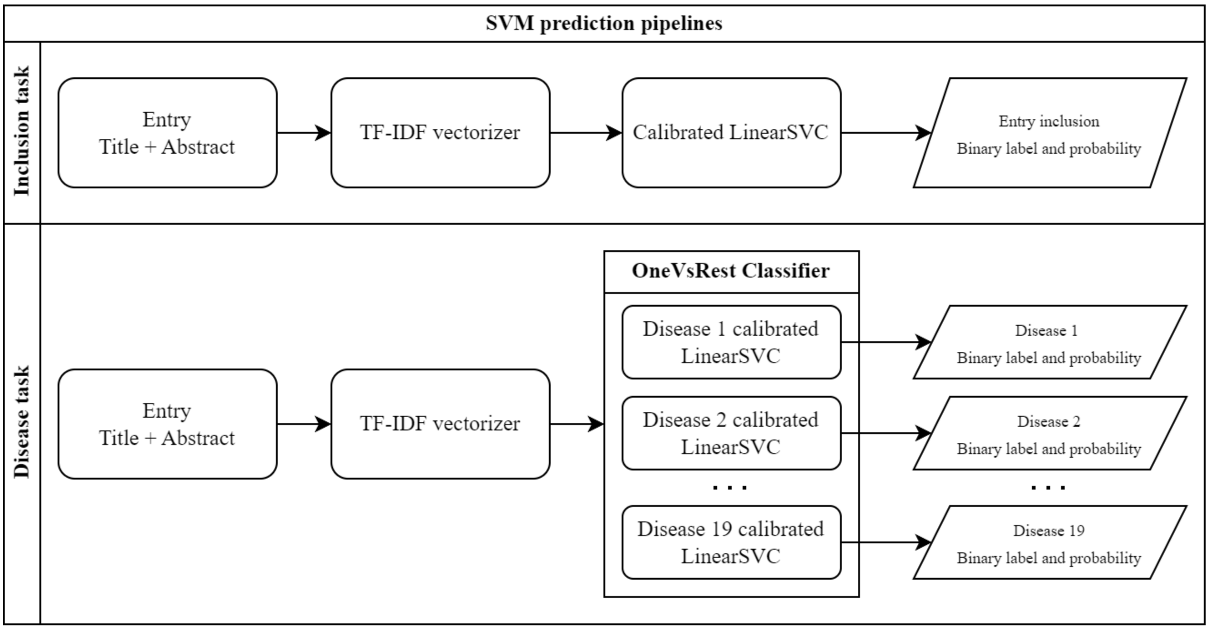


Figure 5: SVM prediction pipelines used in the experiments as flowcharts.

4.3.2 BERT Pipelines

To leverage the contextual capabilities of transformer-based models, a fine-tuned BERT architecture was used. As introduced in Section 2.2, BERT has demonstrated strong performance in biomedical text classification in the literature, owing to its bidirectional attention mechanism and deep language understanding. The models were implemented using the HuggingFace Transformers Python library (Wolf et al., 2020). Text data was preprocessed using a pretrained BERT tokeniser, which converts raw text into token IDs compatible with the model. Since BERT supports a maximum sequence length of 512 tokens, the joined inputs (title + abstract) exceeding this limit were truncated. This affected 4.59% of the sample entries, most of which only marginally exceeded the token threshold (Appendix B.1), and thus were not expected to introduce significant information loss. The model training was performed on a local machine equipped with a GPU using the AdamW optimiser, which is well-suited for training transformer architectures. For the inclusion classification task, the final output layer consisted of two nodes with a softmax activation function, enabling probabilistic interpretation. For the disease classification task, the output layer was extended to 19 nodes - one per disease - with sigmoid activation to allow for multi-label predictions, as each publication may relate to multiple diseases simultaneously. An overview of those pipelines is presented in Figure 6.

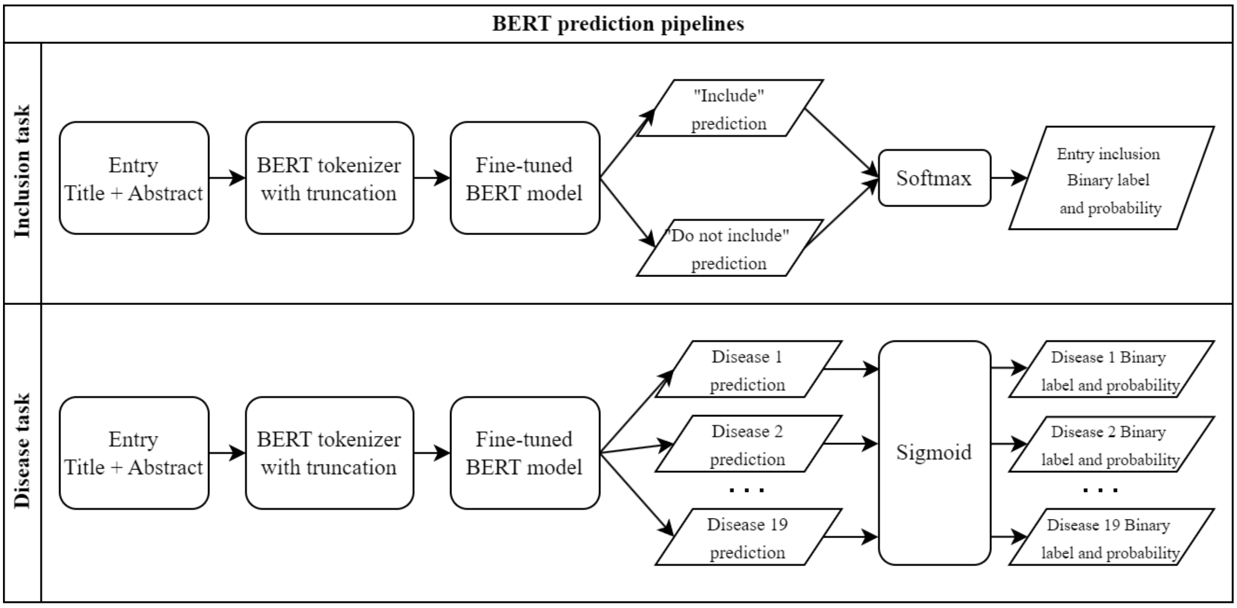


Figure 6: BERT prediction pipelines used in the experiments as flowcharts.

4.3.3 Gemini 2.0 Flash Zero-Shot Pipelines

The third classification approach explored in this thesis leverages a large language model, specifically Gemini 2.0 Flash, developed by Google. This offering of the Gemini family was made to balance performance with inference speed (Google DeepMind, 2024). As a result, it offers lower usage costs, making it a more practical candidate for scalable applications. While a newer version, Gemini 2.5 Flash, was available at the time of writing, it remained in developer preview with strict daily usage limitations, rendering it unsuitable for the requirements of this study.

Gemini 2.0 Flash was used in a zero-shot configuration, meaning that it was not fine-tuned on the task-specific dataset and was not provided with task examples. Instead, the instructions and possible labels were encoded directly into prompts. This flexibility is particularly beneficial for future deployments of the system, as it allows the disease label set to be easily updated without retraining the model. Furthermore, unlike supervised models, the LLM can potentially identify diseases outside the specified taxonomy. The classification tasks were implemented using the google.generativeai Python API. For the inclusion task, the model was prompted to determine whether a given entry was relevant to neglected tropical diseases and output a Yes/No answer. The exact prompt can be seen in Appendix B.2. In the disease classification task, a longer prompt was used, which instructed the model to identify one or more diseases from a predefined list of the 19 NTDs. It also allowed the model to return additional disease names if it deemed the content relevant, offering a mechanism to detect unlisted conditions. A specified JSON format was requested for the answer. The full prompt can be seen in Appendix B.3. As the structure of LLM outputs is not guaranteed to be consistent, a post-processing step was required in both tasks to normalise predictions. Unlike the other approaches described, Gemini 2.0 Flash did not return probability scores, but only categorical predictions. An overview of the model's use in both tasks can be seen in Figure 7.

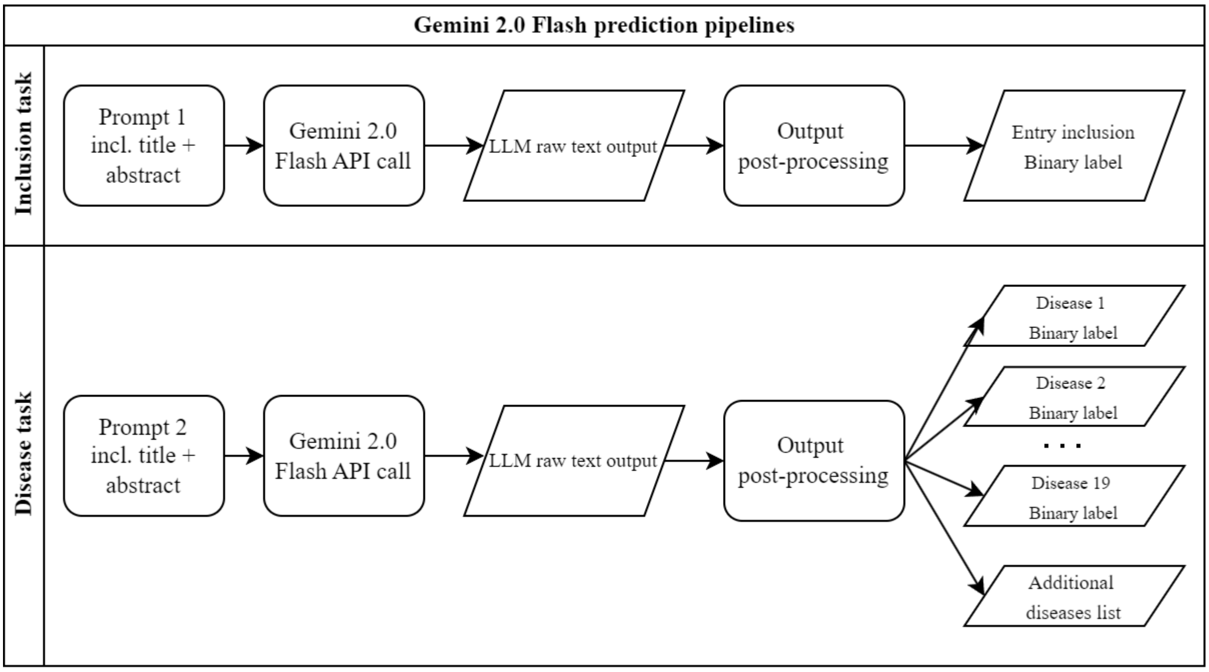


Figure 7: Gemini 2.0 Flash prediction pipelines used in the experiments as flowcharts.

4.3.4 Ensemble Pipelines

In addition to evaluating the individual classification pipelines, this study also investigated whether combining model predictions could enhance performance in the disease classification task. Ensemble methods have been shown in prior research to improve robustness by leveraging the complementary strengths of different model architectures (Sakai & Lam, 2025b; Chen et al., 2021).

Two ensemble strategies were implemented based on the output of the previously described SVM, BERT, and Gemini 2.0 Flash pipelines. In both cases, predictions were aggregated at the individual disease label level for each publication entry. The first method, prediction union, included a disease label in the final output if it was predicted by at least one of the three models. This approach is intended to maximise recall by ensuring that any disease identified by any model is retained. However, this strategy may increase the risk of false positives, thereby potentially reducing overall precision. The second method, majority voting, included a disease label only if it was predicted by at least two of the three models.

This approach aims to balance the benefits of improved recall with a more conservative threshold for inclusion, potentially yielding higher precision than prediction union, while still mitigating the risk of individual model errors.

4.4 Performance metrics

To evaluate the classification performance and practical viability of the models tested in this thesis, a set of performance metrics was used. These metrics allow for assessing not only the predictive accuracy of each approach but also their suitability for practical deployment. For both binary and multi-label classification tasks the following metrics were used:

- Accuracy - represents the overall proportion of correctly classified instances. While it is an intuitive indicator, it can be misleading due to the imbalanced nature of the datasets used.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP are true positives, TN are true negatives, FP are false positives, and FN are false negatives.

- Precision - reflects the proportion of correctly predicted positive labels among all entries classified as positive. In the context of Info Platforms, higher precision means fewer irrelevant publications flagged for inclusion. While important, the cost of lower precision can be partially mitigated in practice by sorting model outputs by prediction confidence, allowing human reviewers to prioritize high-probability entries.

$$Precision = \frac{TP}{TP + FP}$$

- Recall - measures the ability of the model to identify all relevant entries in the dataset. High recall is particularly important in this application, as missing relevant literature could result in important research being overlooked.

$$Recall = \frac{TP}{TP+FN}$$

- F1 score - is the harmonic mean of precision and recall. It provides a balanced measure that accounts for both false positives and false negatives. It is particularly useful for an imbalanced dataset.

$$F1\,Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

For the disease classification task, which is formulated as a multi-label problem, standard metrics need to be adapted to account for multiple labels per entry. To evaluate fairly across both more common and rarer diseases, two types of F1 scores were used:

- Micro-averaged F1 - treats every individual prediction equally, regardless of the label. It adds up the results across the disease labels and then computes the F1 score from those totals. This means that common labels (like leprosy) have a bigger influence on the final score.

$$Micro\,F1 = \frac{2 \times \sum_{D} TP_d}{2 \times \sum_{D} TP_d + \sum_{D} FP_d + \sum_{D} FN_d}$$

  where d is a disease label belonging to the set of labels D.

- Macro-averaged F1 - calculates the F1 score for each label independently and then averages these scores across all labels. It assigns equal weight to each disease, regardless of its frequency, making it useful for assessing how the model performs on rare diseases.

$$Macro\,F1 \;=\; \frac{1}{|D|}\sum_{D} F1\,score_{d}$$

Using both micro and macro F1 scores provides a balanced evaluation - micro F1 shows overall classification effectiveness, while macro F1 shows how well the model generalises across all considered diseases. In addition to the classification metrics, the models were evaluated on their computational efficiency and operational costs with the following metrics:

- Training time - the wall time required to train the supervised models (SVM and BERT), including all preprocessing steps, such as tokenization. It is relevant for assessing the retraining overhead if new labels were to be introduced or new annotated data became available.
- Inference time - how long the model takes to classify a full test set, which impacts the scalability for practical use.
- API usage costs - applies to models accessed through third-party APIs, such as Gemini 2.0 Flash. Every entry classification is associated with monetary costs.

5 Results

5.1 Pipeline results

5.1.1 Entry inclusion task

To assess the effectiveness of different AI models in identifying if a publication is relevant to NTDs, three classification approaches were tested: SVMs and fine-tuned BERT (both trained on the "250k abstracts" train set), and Gemini 2.0 Flash. Their performance was evaluated on the two test datasets described in Section 4.2.1, as well as using only the publication titles included in the "400k mixed" dataset.

All three models demonstrated a very strong performance across test scenarios, with results summarised in Table 3. In the "250k abstracts" set, BERT achieved the highest scores across all evaluation metrics, reaching 0.999 accuracy and an F1 score of 0.993 - indicating near-perfect classification. SVM also performed very well, achieving an F1 score 0.962. Gemini lagged slightly behind in both precision (0.918) and recall (0.820), which is key for the Info Platforms use case. In the more deployment-representative "400k mixed" test set, which includes entries with missing abstracts, performance across all models slightly declined. BERT remained the top performer with an F1 score of 0.948, followed closely by SVMs at 0.941. Gemini again trailed behind, achieving an F1 score of 0.859. The drop in recall was partly caused by the missing abstracts, which is highlighted by the results on the "400k titles only" test set that simulates the minimal-input scenario, providing a more challenging setting. Here, all models showed a further performance drop, with F1 scores in the 0.839 - 0.863 range. Notably, in this setting, the SVM slightly outperformed both BERT and Gemini, possibly due to its smaller reliance on sentence context. The performance gap of Gemini was also smaller, which could be because SVMs and BERT were optimised for longer input during training. These results suggest that all three models are viable options for automated publication inclusion classification, with BERT providing the strongest performance. However, the difference in F1 scores between BERT and SVM is small - particularly in the mixed abstract availability setting - raising questions about whether BERT's computational overhead is justified in the Info Platforms use case. Results in Table 4 show that SVMs are by far the most resource-efficient, requiring only around three minutes to train and ten seconds to classify the test set. In contrast, BERT training took approximately 4.5 hours on a GPU, with inference time of ~12 minutes. This difference would be even larger when using a CPU. Gemini does not require training, but it is constrained by API rate limits and incurs monetary cost. Classifying the test set takes a minimum of 28 minutes and costs around €1.27 at the base Google API pricing tier.

Test set	Model	Accuracy	Precision	Recall	F1 Score
250k abstracts	SVM	0.993	0.971	0.953	0.962
	BERT	**0.999**	**0.992**	**0.993**	**0.993**
	Gemini	0.976	0.918	0.820	0.866
400k mixed	SVM	0.989	0.977	0.907	0.941
	BERT	**0.990**	**0.985**	**0.913**	**0.948**
	Gemini	0.974	0.917	0.807	0.859
400k titles only	SVM	**0.976**	**0.981**	**0.770**	**0.863**
	BERT	0.973	0.980	0.740	0.843
	Gemini	0.971	0.920	0.770	0.839

Table 3: Classification results in the entry inclusion task.

Model	Train time	Inference time	Notes
SVM	~3min	~10sec	Executed on a CPU
BERT	~4h 30min	~12min	Executed on a GPU
Gemini	N/A	~28min*	*API-limited; cost ~€1.27 for the test

Table 4: Training and inference times of models on the "250k abstracts" dataset.

Those comparisons underscore the strong performance-efficiency trade-off of SVMs. While BERT delivers the best results, the marginal gains may not justify the significant increase in compute time. Gemini's performance and ease of use make it attractive for rapid prototyping or small-scale applications, but its lower recall reduces its utility for Info Platforms.

Although all models achieved strong performance across all test sets, a closer inspection of the relatively few misclassifications revealed several patterns that are worth noting. A significant number of errors can be attributed to label noise in the data. Since the negative-class entries were sourced from PubMed and labelled as non-relevant based on the absence of NTD-related MeSH terms, some entries were mislabelled. Manual inspection of entries misclassified by all three models revealed that many of them were actually valid predictions. The ground truth label missed those entries due to missing or incomplete MeSH labelling. Some misclassifications also arose where the titles and abstracts did not clearly indicate a connection to NTDs, while inspecting the publication itself showed it is within scope. For example, one abstract discussed themes of social exclusion and marginalisation without explicitly mentioning NTDs, although the underlying research focused on populations affected by these diseases. Another case involved a medical formulation used across a variety of conditions, where the NTD relevance was not apparent from the text provided to the model. These examples reflect a general limitation of using titles and abstracts alone, which may omit key contextual signals that appear only in the full text. Another trend was specific to the SVM model, which occasionally misclassified publications related to diseases such as malaria or tuberculosis as NTD-relevant. These conditions are not officially included in the NTD taxonomy used for this project, but share thematic and geographic overlap, making them difficult to distinguish without a deeper understanding of context. This limitation was less prominent in the transformer-based models, which were better able to distinguish such borderline cases. Despite the relatively small number of errors, these observations offer useful insights into the limitations of these classification pipelines. They also highlight potential opportunities for improving classification performance further, for instance, by incorporating additional metadata.

5.1.2 Disease classification task

The performance of the models in the disease classification task was evaluated on both test sets described in Section 4.2.2, using two different training set combinations. Table 5 summarises the classification results. The best overall performance was achieved by the

BERT model trained and evaluated on the Combined dataset, reaching a micro-F1 score of 0.968 and a macro-F1 of 0.924. The relatively small gap between these metrics indicates that the model performed well across both frequent and infrequent disease labels. Interestingly, while the BERT trained on the Combined dataset also performed strongly on the Info Platforms test set, the SVMs outperformed it in macro-F1, suggesting better handling of low-resource disease classes. This could be justified by SVMs relying on simpler features which require less data for effective training, giving them an edge in imbalanced classification. Gemini 2.0 Flash, despite not being trained on any task-specific data, also achieved competitive results, especially on the Combined test set. This highlights the potential of LLMs for rapid deployment in biomedical classification tasks, although in this particular case, supervised models still yield higher overall performance.

A key observation across experiments is the significant impact of training data composition on model generalisability. Models trained on the Combined dataset consistently outperformed those trained solely on the Info Platforms data when evaluated on the broader Combined test set. However, the performance difference on the Info Platforms test set was marginal. This highlights how the Info Platforms dataset represents a narrower domain distribution, whereas the Combined set has a higher entry variety. Assuming the MeSH-based annotations in the Combined dataset are sufficiently accurate (see Section 4.2.2), the results emphasise the importance of diverse training data for real-world deployment, especially for less frequently occurring diseases.

The ensemble models (majority voting and prediction union) yielded mixed results. Although ensemble predictions improved over some individual model runs, they did not outperform the strongest single model in any test scenario. This suggests that the individual models often make overlapping errors, limiting the benefits of ensembling. Majority voting yielded slightly better results than prediction union. Full results for all ensemble configurations can be found in Appendix C.19.

Test set	Model	Train Set	Micro-F1	Macro-F1
Info Platforms	SVM	Info Platforms	**0.949**	**0.840**
		Combined	0.919	0.774
	BERT	Info Platforms	0.936	0.643
		Combined	0.940	0.768
	Gemini	N/A	0.902	0.676
	SVM + BERT + Gemini Majority Voting	Info Platforms	0.948	0.795
Combined	SVM	Info Platforms	0.696	0.744
		Combined	0.954	0.902
	BERT	Info Platforms	0.721	0.551
		Combined	**0.968**	**0.924**
	Gemini	N/A	0.940	0.869
	SVM + BERT + Gemini Majority Voting	Combined	0.966	0.922

Table 5: Results in the disease classification task (for all ensemble combinations see Appendix C.19).

Since the performance of the model types was similar, the practical considerations such as computational cost and ease of deployment are especially important. As shown in Table 6, SVMs offer exceptional efficiency. Even though the training and test sets used in this task were much smaller compared to the entry inclusion task, the training and inference times remained similar. This was due to the fact that while many fewer samples were used, the one vs. rest SVM classifier uses 19 models in practice - one for each label. In contrast, the times of BERT were much shorter in this task, requiring half the training time and a third of the inference time. This is due to the ability of BERT to output many labels using the same neural network. However, even with this change, the SVMs remain around 45 times faster.

While BERT delivered the highest F1 scores, the marginal performance gain over SVMs may not justify the substantial increase in resource requirements. Gemini 2.0 Flash provides a viable alternative to both models, offering ease of use and strong performance without the need for training. Its zero-shot capability allows for straightforward expansion of the disease taxonomy by simply adding labels to the prompt. Although API limitations result in longer inference times (8.5 minutes for the Combined test set) and a small cost, these constraints are relatively minor. The Combined test set contained 16,574 entries and classifying them amounted to around €1.77. The entire library of Info Platforms consists of 39,070 unique entries, implying the monthly classification costs would be negligible.

Model	Train time	Inference time	Notes
SVM	~3min	~6sec	Executed on a CPU
BERT	~2h 12min	~4.5min	Executed on a GPU
Gemini	N/A	~8.5min*	*API-limited; cost ~€1.77 for the test

Table 6: Training and inference times of models in the disease classification task on the "Combined" dataset.

To understand the limitations of the tested pipelines, the misclassified entries were analyzed. Across all models, errors frequently occurred when the input (title and abstract) confirmed NTD relevance but failed to explicitly mention specific diseases. These ambiguous cases were particularly problematic in multi-label entries, where the likelihood of at least one incorrect prediction increases with the number of associated diseases (Figure 8). Interestingly, Gemini 2.0 Flash demonstrated slightly better handling of multi-disease entries compared to SVMs and BERT. Beyond six labels, however, all models consistently failed to produce correct predictions. Nevertheless, given that 97.6% of entries in the dataset have only one disease label, this limitation is of limited practical concern. A future improvement could involve introducing a generic "Multiple NTDs" category for such cases.

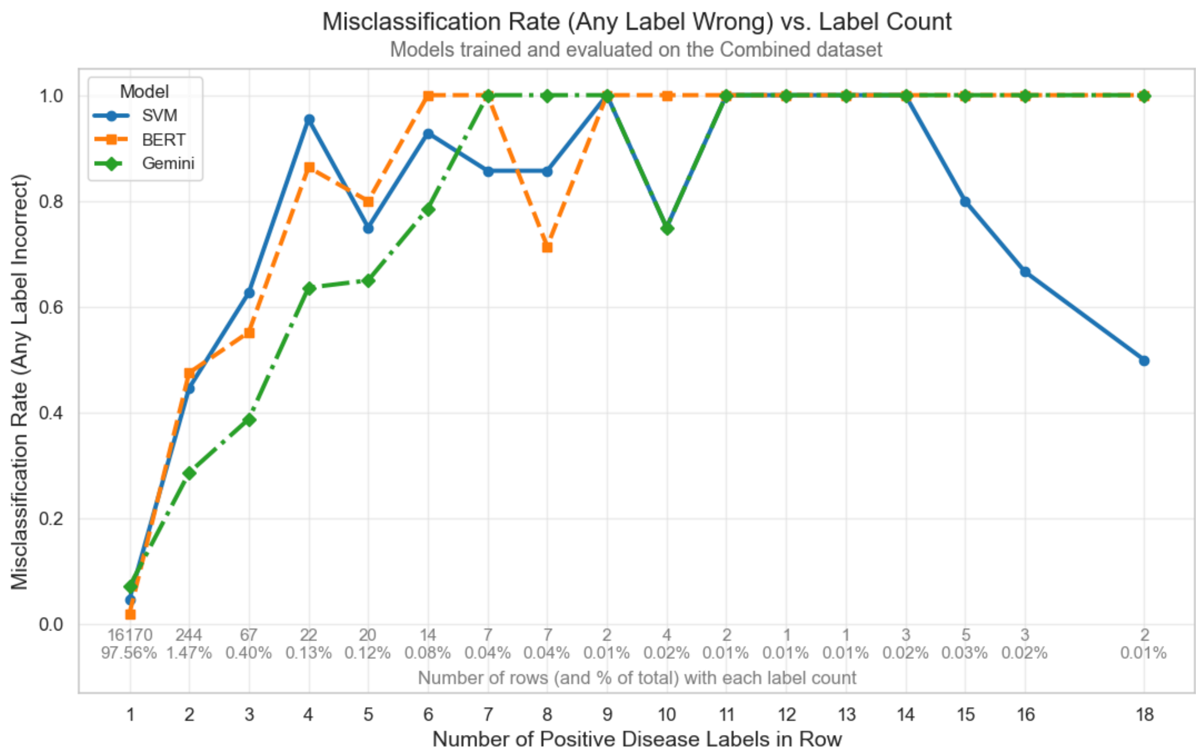


Figure 8: Performance of the models tested in the disease classification task on the "Combined" test set in comparison to the total number of correct disease labels of the entry.

The performance of the models per disease was also analysed to identify more difficult labels. Figure 9 presents precision and recall per disease for the models trained and evaluated on the Combined dataset. While performance was generally strong, several diseases consistently exhibited lower scores across all models (seen as lighter coloured rows in Figure 9), notably dracunculiasis, lymphatic filariasis, podoconiosis, and yaws. While sample size alone does not directly determine classification performance - as demonstrated by diseases such as Chromoblastomycosis, which achieved strong results despite relatively few samples - all of the underperforming diseases had limited representation in the dataset. This scarcity likely amplified the impact of individual misclassifications on the overall evaluation metrics. The impact of the split-size increase from the Info Platforms set to the Combined set was not consistent across diseases.

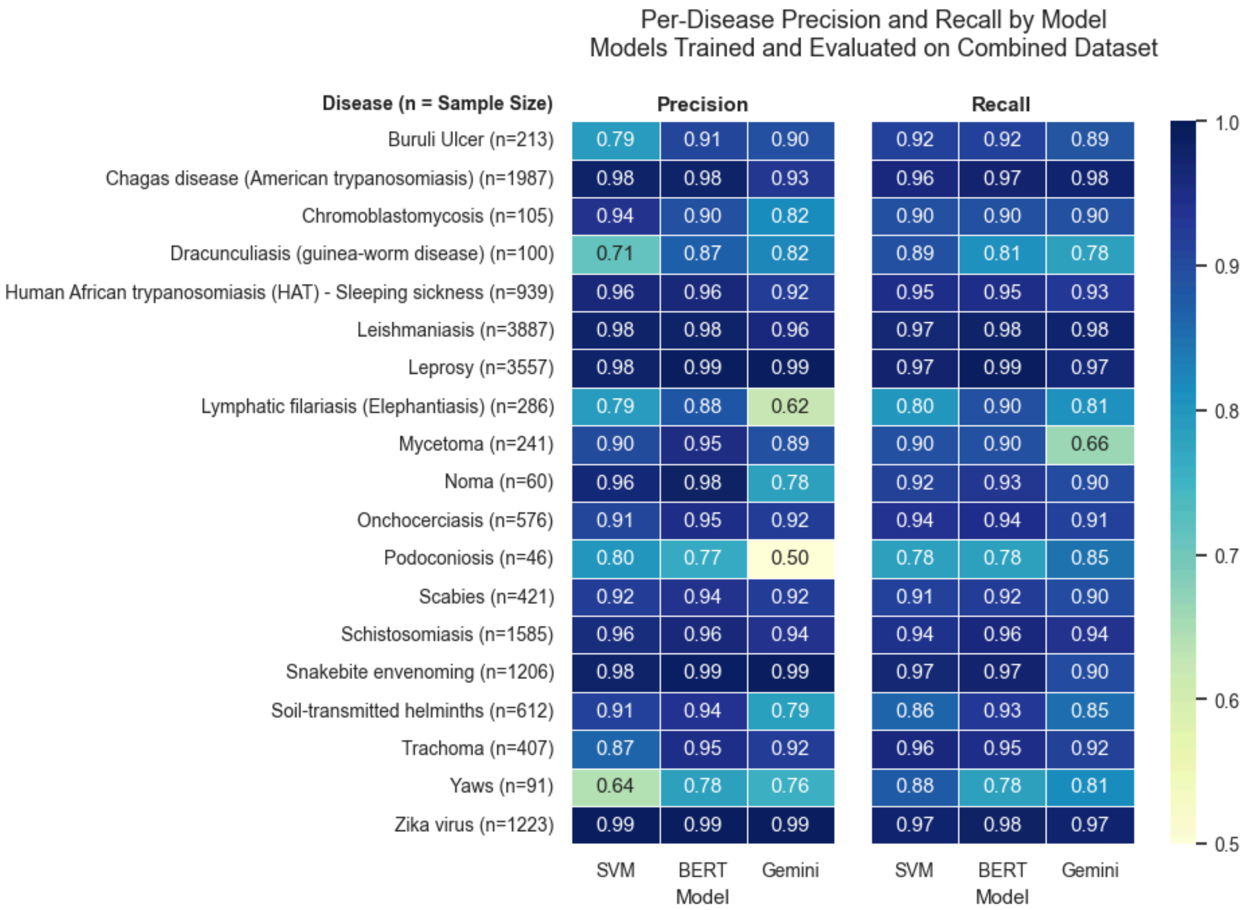


Figure 9: Precision and recall of the models tested in the disease classification task on the "Combined" test across the 19 NTDs assessed.

A common pattern among the four lowest performing diseases was their association with publications that discussed multiple NTDs simultaneously. Although this issue affected many diseases, its consequences were more severe for those with smaller support, where even a few ambiguous examples significantly reduced precision and recall. Another source of error was label inconsistency, including both incomplete MeSH annotations and occasional inaccuracies in expert labelling. In several instances, the abstract clearly referenced a disease present in the classification taxonomy, yet the corresponding ground truth label was incorrect. While these cases were relatively infrequent, they introduced label noise that negatively impacted model scores by penalising accurate predictions. A particularly prominent confusion pattern emerged between podoconiosis and lymphatic filariasis. These two diseases share overlapping symptoms and similar geographical prevalence, which likely contributed to their frequent misclassification. Overall, these findings indicate that a substantial portion of the observed model errors stem from data-related limitations rather than inherent model deficiencies. Addressing label ambiguity could further improve model performance.

5.2 Historical data analysis

To assess the broader applicability and value of the developed classification pipelines, the trained SVM models for both inclusion and disease classification tasks were applied to the entire PubMed 2025 Annual Baseline. Using the inclusion classification model on the 36.5 million entries of that dataset, 188,304 entries were predicted to be within the scope of the Info Platforms. Those predictions included 26,751 entries already present in the Info Platforms, identified by title and DOI matching. This result represents a recall of 91.4% relative to the 29,267 known matched entries in the PubMed 2025 Annual Baseline (Section 4.1.2). Notably, this result was achieved using a default probability threshold of 0.5 for positive classification. While lowering the threshold could increase the recall further, it would likely reduce precision - highlighting a potential area for future calibration depending on the user's tolerance for false positives.

The disease classification model was then applied to the subset of entries predicted to be relevant. This allowed for an estimation of the potential expansion in coverage for each of the 19 target NTDs. The predicted counts per disease are summarised in Figure 10, with a complete breakdown in Appendix C.20. For instance, the number of articles predicted to be relevant to Leishmaniasis suggests a 28.6-fold increase relative to the current coverage in the Info Platforms. The smallest increase was projected for Podoconiosis (1.4-fold), indicating that current platform coverage for this disease is already relatively comprehensive. Overall, the historical analysis demonstrates the large potential for using AI-powered pipelines for identifying previously overlooked publications, improving the comprehensiveness of the Info Platforms.

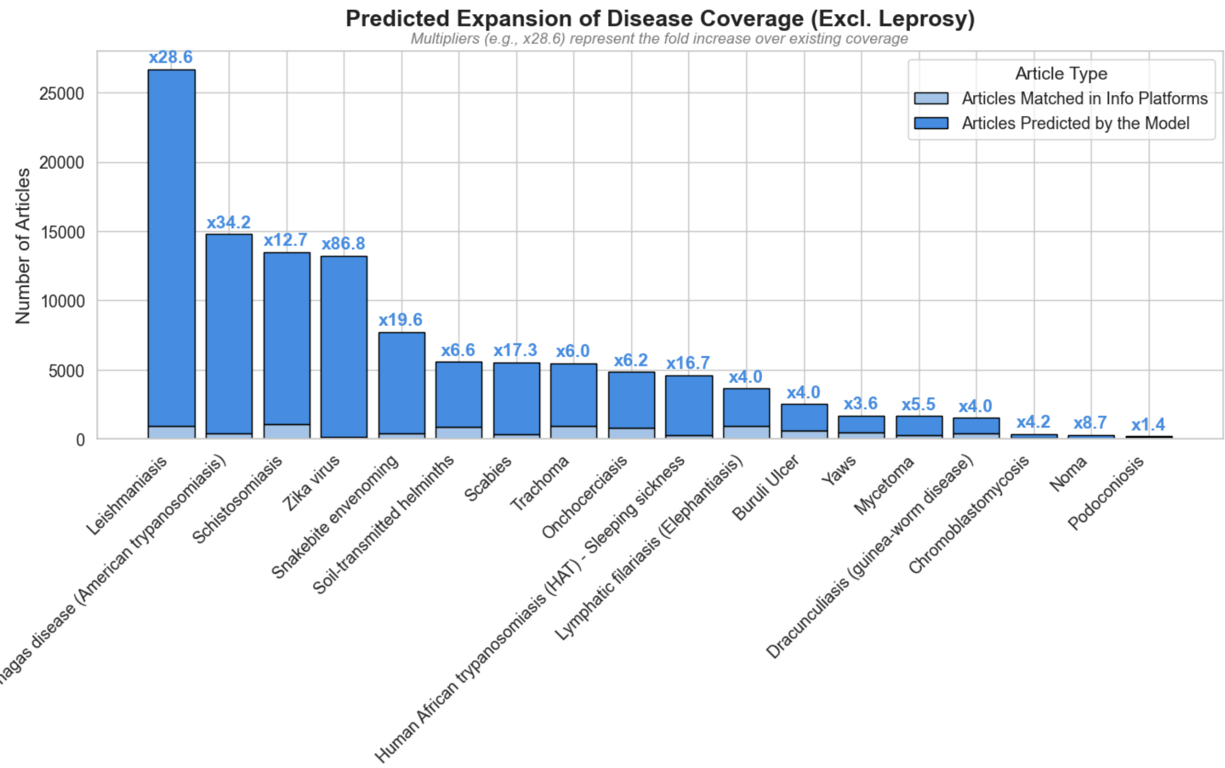


Figure 10: Predicted numbers of articles relevant to Info Platforms per disease in the PubMed 2025 Annual Baseline dataset and the growth factor compared to current Info Platforms coverage using SVM models. The figure excludes Leprosy due to a large scale mismatch. Leprosy had 57,160 identified articles, 22,648 out of these were matched with Info Platforms, representing a 2.5-fold increase. (Appendix C.20)

6 Limitations and future research

While the results of this thesis demonstrate that AI-powered pipelines can effectively support the curation of NTD-related literature, several limitations must be acknowledged. First, the disease classification task was conducted exclusively on entries containing abstracts, which provide essential contextual information for accurate prediction. Consequently, the models were not evaluated on records without abstracts, limiting their applicability to minimal-content scenarios. Furthermore, the training data relied on MeSH terms to construct ground truth labels. Although the experiments have shown that MeSH annotations can improve classification accuracy, they are sometimes incomplete or inconsistent for rare diseases, including many NTDs (Rei et al., 2024). For instance, there are no MeSH terms for podoconiosis, hence, the number of samples in the Combined disease dataset could not be enhanced over the Info Platforms inclusions. The annotations are also rarely updated retroactively when new categories are introduced (Fatehi & Gray, 2013). This means there is no guarantee that the datasets fully captured all relevant literature, particularly in evolving disease areas. Another important consideration is the linguistic composition of the training data. While a small number of entries in other languages were included, the overwhelming majority of the data was in English. Given that English dominates global biomedical publishing, this is unlikely to impair overall model utility. However, it may hinder performance on local-language sources - especially relevant for diseases that are geographically concentrated in non-English-speaking regions. Lastly, although BERT models performed well during experimentation, they were trained and tested on a GPU, which may not be available in all deployment contexts. Running such models on a CPU alone would result in significantly slower inference.

Looking ahead, several promising directions for future research emerge. One avenue is to introduce a generic NTD label, applicable when a publication discusses multiple diseases without clearly focusing on individual ones. Exploring larger or more specialised models, such as GPT-4o or biomedical-tuned BERT variants, could also enhance classification performance. Likewise, future work could experiment with alternative ensemble configurations, testing different combinations or weightings of models to better balance precision and recall. Beyond disease classification, extending the system to more complex tagging tasks - such as identifying topic areas or geographic regions - would further support the automation efforts of Infolep and InfoNTD. These tags are often less explicitly mentioned in text and more inconsistently applied, posing greater challenges. However, success in this area could substantially reduce manual curation effort. Additionally, while the present study focused exclusively on scientific publications from structured repositories, the platforms themselves aim to include a broader range of materials, including news articles and training courses. These content types were not addressed in this research and may require the development of tailored data ingestion and classification strategies before they can be integrated effectively. Future research could explore alternative data sources, such as scraping or integrating web-based content relevant to NTDs.

7 Deliverable

To enable non-technical stakeholders at the Infolep and InfoNTD platforms to explore the utility of AI-based content management pipelines, a proof-of-concept tool was developed as part of this thesis. The primary objective of this tool is to provide an accessible interface through which users can assess the practical implications of automated classification systems within their existing workflows, prior to considering investment in a fully integrated solution. The tool features a graphical user interface (GUI), shown in Figure 11, designed to configure the automated pipeline. Users can specify the desired time frame for scanning newly published literature, select the sources from which to retrieve publication data via APIs (as described in Section 4.1.3), and choose which inclusion models to apply, including configurable probability thresholds. Additionally, users may select from available disease classification models to generate relevant disease predictions.

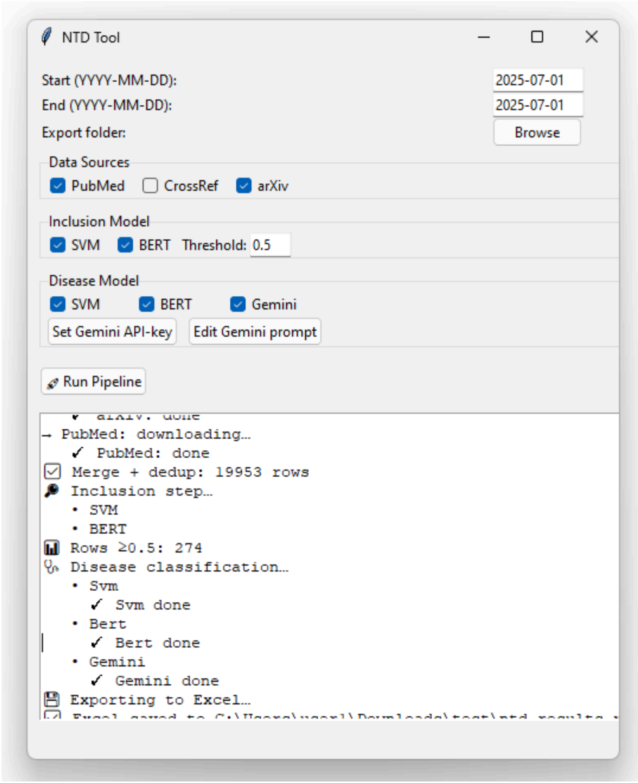


Figure 11: Graphical User Interface of the proof of concept tool developed for the Info Platforms.

Upon initiating the pipeline by selecting the *Run pipeline* button, the process outlined in Figure 12 is executed. The tool first aggregates and deduplicates publications retrieved from the selected sources, consolidating them into a unified dataset. Next, the inclusion models are applied to this dataset, filtering it down to only those publications that are predicted to be within the scope of NTDs by at least one model, according to the specified threshold criteria. Subsequently, disease classification models are run on the filtered subset, and their predictions are appended to the publication records. The final output is an organized Microsoft Excel file (Figure 13) containing the metadata associated with each publication along with the predictions produced by each model. This file is intended to support human review and manual integration of relevant publications into the Infolep and InfoNTD platforms. To further assist end-users in selecting appropriate models for their use case, Table 7 compares the tested models - SVMs, fine-tuned BERT, and Gemini 2.0 Flash - based on their performance characteristics, interpretability, and operational considerations.

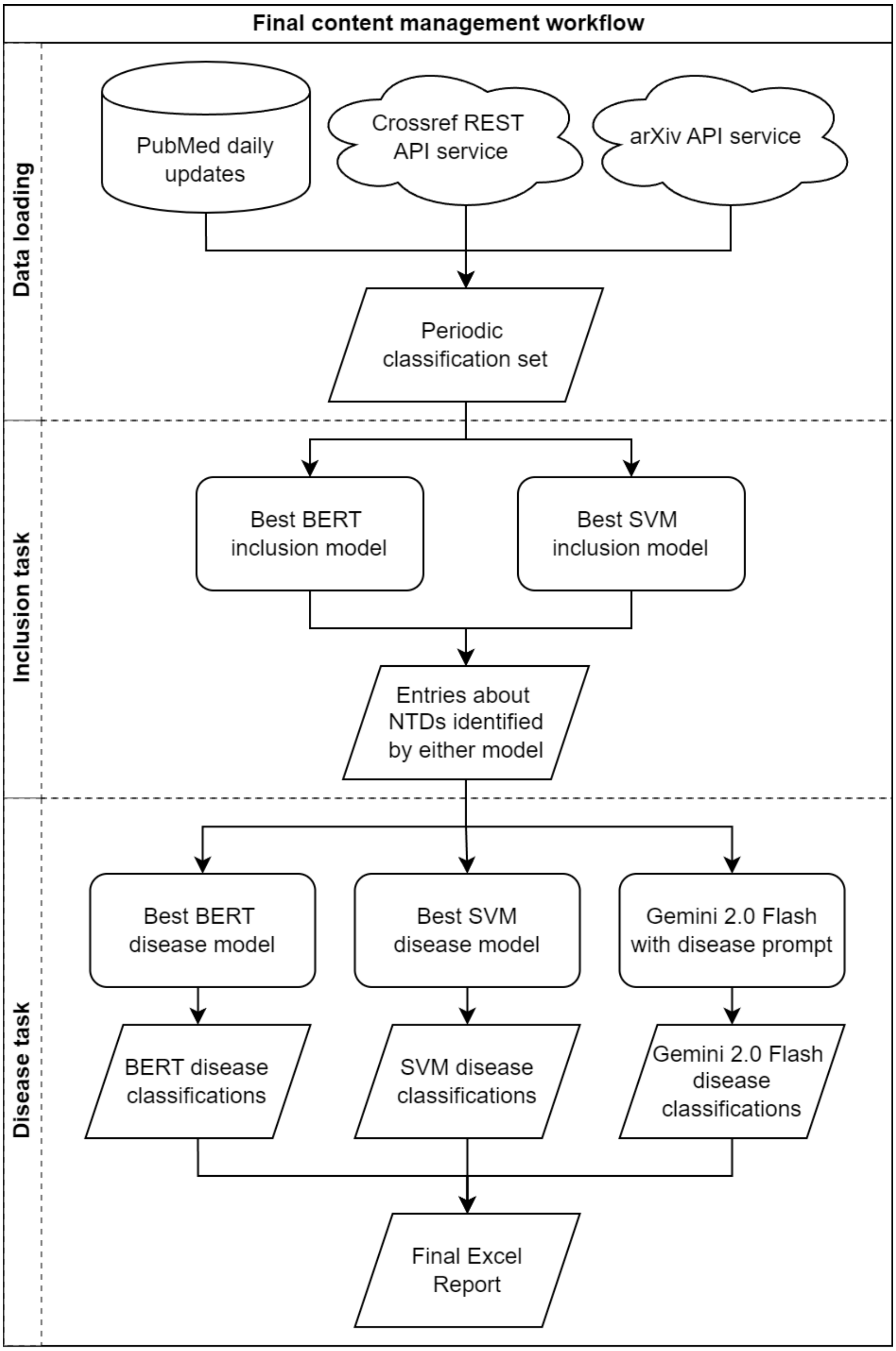
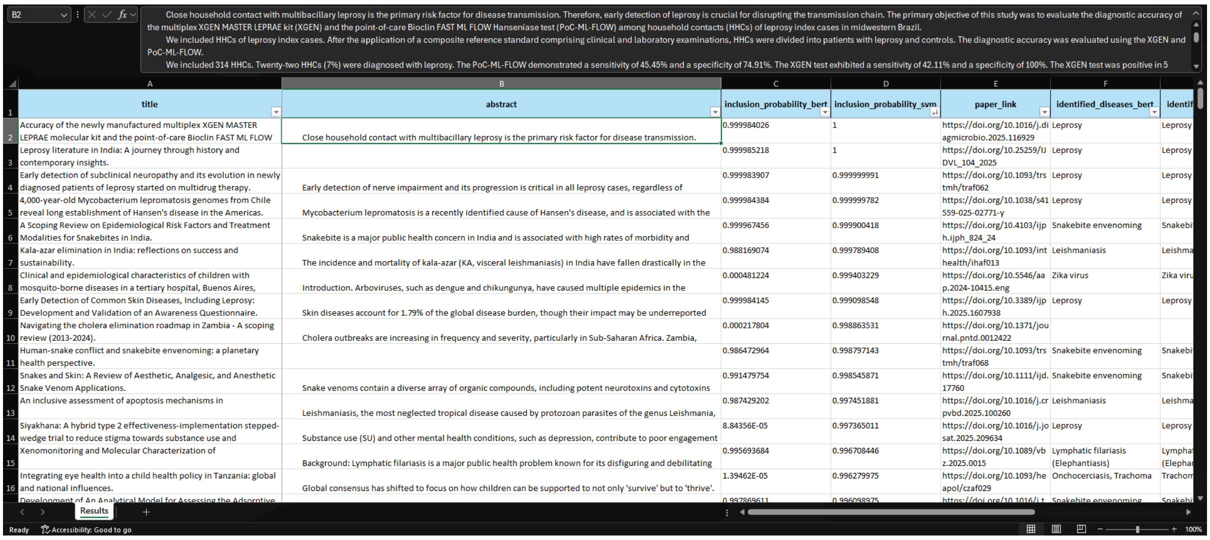


Figure 12: Publication scanning pipeline deployed in the proof-of-concept tool developed for the Info Platforms.



	A	B	C	D	E	F	
1	title	abstract	inclusion_probability_bert	inclusion_probability_svm	paper_link	identified_diseases_bert	identif
2	Accuracy of the newly manufactured multiplex XGEN MASTER LEPRAE molecular kit and the point-of-care Bioclin FAST ML FLOW	Close household contact with multibacillary leprosy is the primary risk factor for disease transmission.	0.999984026	1	https://doi.org/10.1016/j.diagmicrobio.2025.116929	Leprosy	Leprosy
3	Leprosy literature in India: A journey through history and contemporary insights.		0.999985218	1	https://doi.org/10.25259/IJDVL_104_2025	Leprosy	Leprosy
4	Early detection of subclinical neuropathy and its evolution in newly diagnosed patients of leprosy started on multidrug therapy.	Early detection of nerve impairment and its progression is critical in all leprosy cases, regardless of	0.999983907	0.999999991	https://doi.org/10.1093/trstmh/traf062	Leprosy	Leprosy
5	4,000-year-old Mycobacterium lepromatosis genomes from Chile reveal long establishment of Hansen's disease in the Americas.	Mycobacterium lepromatosis is a recently identified cause of Hansen's disease, and is associated with the	0.999984384	0.999999782	https://doi.org/10.1038/s41559-025-02771-y	Leprosy	Leprosy
6	A Scoping Review on Epidemiological Risk Factors and Treatment Modalities for Snakebites in India.	Snakebite is a major public health concern in India and is associated with high rates of morbidity and	0.999967456	0.999900418	https://doi.org/10.4103/ijph.ijph_824_24	Snakebite envenoming	Snakebi
7	Kala-azar elimination in India: reflections on success and sustainability.	The incidence and mortality of kala-azar (KA, visceral leishmaniasis) in India have fallen drastically in the	0.988169074	0.999789408	https://doi.org/10.1093/inthealth/ihaf013	Leishmaniasis	Leishma
8	Clinical and epidemiological characteristics of children with mosquito-borne diseases in a tertiary hospital, Buenos Aires,	Introduction. Arboviruses, such as dengue and chikungunya, have caused multiple epidemics in the	0.000481224	0.999403229	https://doi.org/10.5546/aap.2024-10415.eng	Zika virus	Zika viru
9	Early Detection of Common Skin Diseases, Including Leprosy: Development and Validation of an Awareness Questionnaire.	Skin diseases account for 1.79% of the global disease burden, though their impact may be underreported	0.999984145	0.999098548	https://doi.org/10.3389/ijph.2025.1607938	Leprosy	Leprosy
10	Navigating the cholera elimination roadmap in Zambia - A scoping review (2013-2024).	Cholera outbreaks are increasing in frequency and severity, particularly in Sub-Saharan Africa. Zambia,	0.000217804	0.998863531	https://doi.org/10.1371/journal.pntd.0012422		
11	Human-snake conflict and snakebite envenoming: a planetary health perspective.		0.986472964	0.998797143	https://doi.org/10.1093/trstmh/traf068	Snakebite envenoming	Snakebi
12	Snakes and Skin: A Review of Aesthetic, Analgesic, and Anesthetic Snake Venom Applications.	Snake venoms contain a diverse array of organic compounds, including potent neurotoxins and cytotoxins	0.991479754	0.998545871	https://doi.org/10.1111/ijd.17760	Snakebite envenoming	Snakebi
13	An inclusive assessment of apoptosis mechanisms in	Leishmaniasis, the most neglected tropical disease caused by protozoan parasites of the genus Leishmania,	0.987429202	0.997451881	https://doi.org/10.1016/j.crpvbd.2025.100260	Leishmaniasis	Leishma
14	Siyakhana: A hybrid type 2 effectiveness-implementation stepped-wedge trial to reduce stigma towards substance use and	Substance use (SU) and other mental health conditions, such as depression, contribute to poor engagement	8.84356E-05	0.997365011	https://doi.org/10.1016/j.josat.2025.209634	Leprosy	Leprosy
15	Xenomonitoring and Molecular Characterization of	Background: Lymphatic filariasis is a major public health problem known for its disfiguring and debilitating	0.995693684	0.996708446	https://doi.org/10.1089/vbz.2025.0015	Lymphatic filariasis (Elephantiasis)	Lympha (Elepha
16	Integrating eye health into a child health policy in Tanzania: global and national influences.	Global consensus has shifted to focus on how children can be supported to not only 'survive' but to 'thrive'.	1.39462E-05	0.996279975	https://doi.org/10.1093/heapol/czaf029	Onchocerciasis, Trachoma	Trachom

Figure 13: Microsoft Excel export of the model predictions with publication metadata to be used in the proof of concept content management workflow of the Info Platforms.

Model	Benefits	Drawbacks
SVMs	• Very fast • Strong performance • Probability scores	• Limited contextual understanding • Requires retraining for new labels
BERT	• Best performance • Good contextual understanding • Probability scores	• Slow • Requires significant compute (GPU recommended) • Requires retraining for new labels
Gemini 2.0 Flash	• No training needed • Easy to update disease taxonomy • Good performance out of the box	• API rate limits and costs • No probability scores • Slightly worse performance compared to BERT and SVMs

Table 7: Summary of the benefits and drawbacks of tested models for practical use.

8 Conclusions

Biomedical literature is expanding rapidly. Scalable, accurate curation methods are therefore critical for knowledge platforms such as Infolep and InfoNTD. The research conducted for this thesis highlights significant opportunities offered by artificial intelligence to enhance the content management pipelines and reduce the manual workload.

To facilitate the experiments, four datasets were constructed by integrating expert-annotated records from the Info Platforms with extensive biomedical literature from PubMed, enriched with targeted MeSH terms. This hybrid approach enabled the creation of comprehensive experimental scenarios, providing both negative-class samples for the publication inclusion task and an enriched dataset to address small label support in the disease classification task. The evaluation of different modelling techniques highlighted relatively minor differences between their performance. Fine-tuned BERT transformer models achieved the best overall classification results due to their advanced contextual understanding. However, their modest performance advantage comes with significant compute requirements, which limits practical deployment. Support vector machines, on the other hand, offered comparable performance with exceptional computational efficiency, making them particularly suitable for the Info Platforms use case. The zero-shot Gemini 2.0 Flash large language model provided valuable flexibility in the taxonomy and was easy to implement, but yielded slightly lower results in classification. Ensemble approaches did not provide improvements over the best individual models, indicating that their errors were not complementary. The analysis of misclassified entries revealed challenges such as ambiguity in abstracts, ground-truth annotation inconsistencies, and difficulties in distinguishing between diseases with similar characteristics. Future work could mitigate these issues by adding broader classification categories. Historical data analysis using the SVMs demonstrated the value of automating literature monitoring and classification, revealing substantial potential to improve the coverage across many neglected tropical diseases. Finally, the proof-of-concept tool for literature retrieval and classification showed how such AI-powered content management pipelines could be deployed in practice.

Based on these findings, the recommendation for Infolep and InfoNTD is to begin their AI implementation by adopting the SVM-based pipelines into their curation process. Their efficiency and ease of deployment make them a good starting point for automating publication screening without requiring significant technical infrastructure. To enhance precision in the longer term, a BERT-based model could be used. Large language models, such as Gemini 2.0 Flash, could be employed for classification tasks where flexible taxonomies are required. This adaptive integration of AI can help the Info Platforms to not only scale their operations but also more effectively respond to the dynamic landscape of global health research on neglected tropical diseases.

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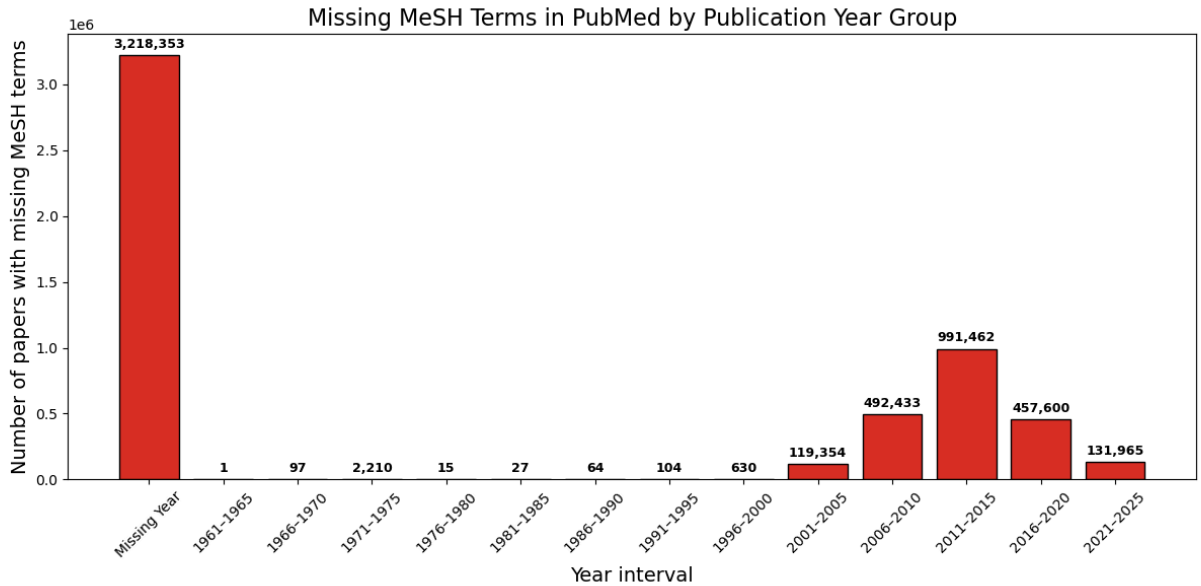
Appendix

Appendix A - Data

Appendix A.1 - Number of papers with missing MeSH terms per year.

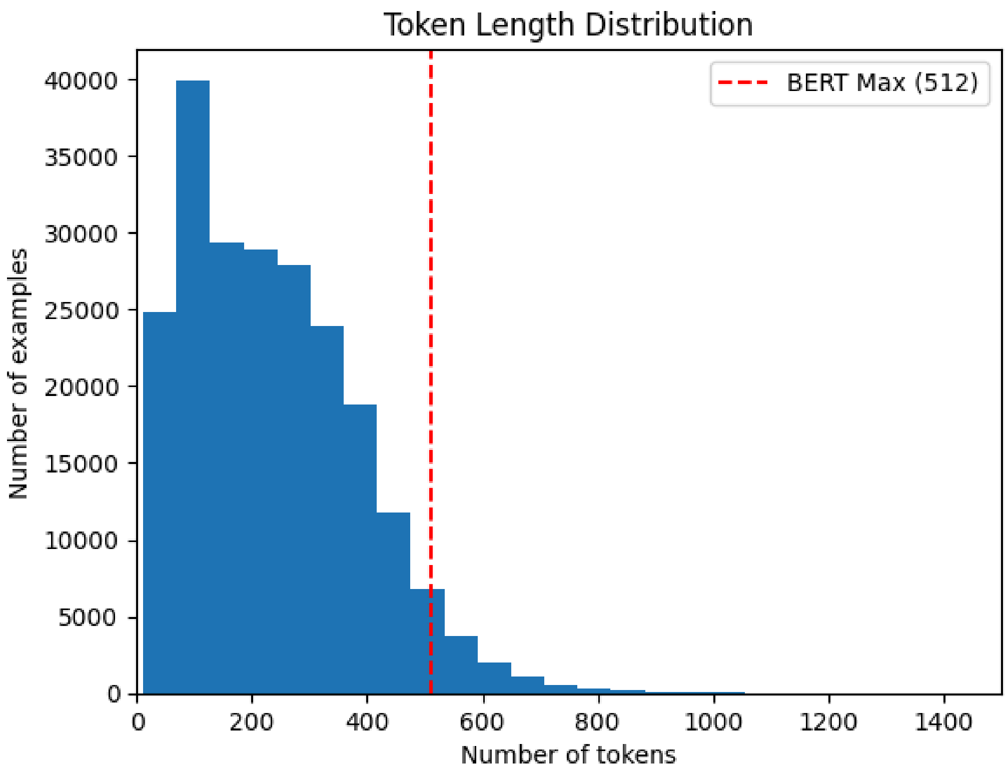


Appendix A.2 - MeSH terms assigned target class 1

"Schistosomiasis","Lymphatic Filariasis", "Elephantiasis", "Leprosy", "Leprosy, Tuberculoid", "Leprosy, Lepromatous", "Leprosy, Borderline", "Leprosy, Paucibacillary", "Leprosy, Multibacillary", "Soil-Transmitted Helminth Infections", "Ascaris lumbricoides", "Trichuris trichiura", "Hookworm Infections", "Leishmaniasis", "Leishmaniasis, Cutaneous", "Leishmaniasis, Visceral", "Onchocerciasis", "Trachoma", "Chagas Disease", "American Trypanosomiasis", "Buruli Ulcer", "Snake Bites", "Snake Venoms", "Trypanosomiasis, African", "Sleeping Sickness", "Yaws", "Mycetoma", "Eumycetoma", "Actinomycetoma", "Podoconiosis", "Dracunculiasis", "Guinea Worm Disease", "Scabies", "Zika Virus Infection", "Chromoblastomycosis", "Noma"

Appendix B - Pipelines

Appendix B.1 - Token Length distribution when using a BERT tokeniser on the title + abstract text fields from the 250k abstracts dataset.



Appendix B.2 - Gemini prompt used in the entry inclusion task:

You are an expert in tropical diseases. Given the following medical research paper's title and abstract, answer only with 'Yes' or 'No':

{title + abstract}

Is this paper about neglected tropical diseases?

Appendix B.3 - Gemini prompt used in the disease classification task:

You are an expert in tropical diseases. Given the following medical research paper's title and abstract, identify which diseases are discussed. You have the following list of known neglected tropical diseases (NTDs):

['Buruli Ulcer',
'Chagas disease (American trypanosomiasis)',
'Chromoblastomycosis',
'Dracunculiasis (guinea-worm disease)',
'Human African trypanosomiasis (HAT) - Sleeping sickness',
'Leishmaniasis',
'Leprosy',
'Lymphatic filariasis (Elephantiasis)',
'Mycetoma',
'Noma',
'Onchocerciasis',
'Podoconiosis',
'Scabies',
'Schistosomiasis',
'Snakebite envenoming',
'Soil-transmitted helminths',
'Trachoma',
'Yaws',
'Zika virus']

Use the exact names from the list when applicable. If the paper discusses diseases not in the list, include them too, using their commonly accepted names.

Return your answer as a JSON object in the following format:

{"diseases": ["Disease 1", "Disease 2", ...]}

Here is the publication:

{title + abstract}

Appendix C - Results

Appendix C.1 - Disease multilabel classification

Model: SVM **Train: Info Platforms** **Test: Info Platforms**

	precision	recall	f1-score	support
Buruli Ulcer	0.891	0.885	0.888	130
Chagas disease (American trypanosomiasis)	0.846	0.750	0.795	88
Chromoblastomycosis	0.778	0.583	0.667	12
Dracunculiasis (guinea-worm disease)	0.806	0.806	0.806	31
Human African trypanosomiasis (HAT) - Sleeping sickness	0.690	0.755	0.721	53
Leishmaniasis	0.800	0.852	0.825	169
Leprosy	0.992	0.991	0.992	3495
Lymphatic filariasis (Elephantiasis)	0.884	0.876	0.880	226
Mycetoma	0.814	0.814	0.814	43
Noma	1.000	1.000	1.000	7
Onchocerciasis	0.880	0.798	0.837	119
Podoconiosis	0.909	0.870	0.889	46
Scabies	0.846	0.611	0.710	36
Schistosomiasis	0.933	0.863	0.897	226
Snakebite envenoming	0.914	0.928	0.921	69
Soil-transmitted helminths	0.907	0.865	0.885	192
Trachoma	0.897	0.852	0.874	122
Yaws	0.809	0.809	0.809	47
Zika virus	0.812	0.684	0.743	19
micro avg	0.954	0.943	0.949	5130
macro avg	0.864	0.821	0.840	5130
weighted avg	0.954	0.943	0.948	5130
samples avg	0.974	0.976	0.972	5130

Appendix C.2 - Disease multilabel classification

Model: SVM **Train: Info Platforms** **Test: Combined**

	precision	recall	f1-score	support
Buruli Ulcer	0.923	0.906	0.915	213
Chagas disease (American trypanosomiasis)	0.980	0.682	0.804	1987
Chromoblastomycosis	0.959	0.448	0.610	105
Dracunculiasis (guinea-worm disease)	0.887	0.550	0.679	100
Human African trypanosomiasis (HAT) - Sleeping sickness	0.933	0.444	0.602	939
Leishmaniasis	0.979	0.620	0.759	3887
Leprosy	0.389	0.992	0.558	3557
Lymphatic filariasis (Elephantiasis)	0.779	0.738	0.758	286
Mycetoma	0.887	0.456	0.603	241
Noma	1.000	0.700	0.824	60
Onchocerciasis	0.952	0.648	0.771	576
Podoconiosis	0.755	0.870	0.808	46
Scabies	0.981	0.482	0.646	421
Schistosomiasis	0.968	0.676	0.796	1585
Snakebite envenoming	0.991	0.611	0.756	1206
Soil-transmitted helminths	0.869	0.552	0.675	612
Trachoma	0.954	0.811	0.876	407
Yaws	0.861	0.747	0.800	91
Zika virus	0.997	0.820	0.900	1223
micro avg	0.679	0.714	0.696	17542
macro avg	0.897	0.671	0.744	17542
weighted avg	0.846	0.714	0.725	17542
samples avg	0.661	0.715	0.678	17542

Appendix C.3 - Disease multilabel classification

Model: SVM **Train: Combined** **Test: Info Platforms**

	precision	recall	f1-score	support
Buruli Ulcer	0.712	0.892	0.792	130
Chagas disease (American trypanosomiasis)	0.815	0.750	0.781	88
Chromoblastomycosis	0.583	0.583	0.583	12
Dracunculiasis (guinea-worm disease)	0.467	0.903	0.615	31
Human African trypanosomiasis (HAT) - Sleeping sickness	0.760	0.717	0.738	53
Leishmaniasis	0.746	0.817	0.780	169
Leprosy	0.987	0.974	0.981	3495
Lymphatic filariasis (Elephantiasis)	0.838	0.850	0.844	226
Mycetoma	0.686	0.814	0.745	43
Noma	1.000	1.000	1.000	7
Onchocerciasis	0.694	0.857	0.767	119
Podoconiosis	0.878	0.783	0.828	46
Scabies	0.418	0.639	0.505	36
Schistosomiasis	0.862	0.885	0.873	226
Snakebite envenoming	0.863	0.913	0.887	69
Soil-transmitted helminths	0.902	0.818	0.858	192
Trachoma	0.696	0.918	0.792	122
Yaws	0.488	0.851	0.620	47
Zika virus	0.700	0.737	0.718	19
micro avg	0.906	0.932	0.919	5130
macro avg	0.742	0.826	0.774	5130
weighted avg	0.919	0.932	0.923	5130
samples avg	0.950	0.963	0.951	5130

Appendix C.4 - Disease multilabel classification

Model: SVM **Train: Combined** **Test: Combined**

	precision	recall	f1-score	support
Buruli Ulcer	0.793	0.915	0.850	213
Chagas disease (American trypanosomiasis)	0.978	0.957	0.968	1987
Chromoblastomycosis	0.940	0.895	0.917	105
Dracunculiasis (guinea-worm disease)	0.712	0.890	0.791	100
Human African trypanosomiasis (HAT) - Sleeping sickness	0.961	0.947	0.954	939
Leishmaniasis	0.980	0.969	0.975	3887
Leprosy	0.979	0.972	0.975	3557
Lymphatic filariasis (Elephantiasis)	0.789	0.797	0.793	286
Mycetoma	0.904	0.896	0.900	241
Noma	0.965	0.917	0.940	60
Onchocerciasis	0.909	0.939	0.924	576
Podoconiosis	0.800	0.783	0.791	46
Scabies	0.919	0.912	0.915	421
Schistosomiasis	0.955	0.942	0.949	1585
Snakebite envenoming	0.978	0.968	0.973	1206
Soil-transmitted helminths	0.912	0.864	0.888	612
Trachoma	0.867	0.958	0.910	407
Yaws	0.640	0.879	0.741	91
Zika virus	0.989	0.975	0.982	1223
micro avg	0.956	0.952	0.954	17542
macro avg	0.893	0.915	0.902	17542
weighted avg	0.958	0.952	0.955	17542
samples avg	0.956	0.964	0.958	17542

Appendix C.5 - Disease multilabel classification

Model: BERT **Train: Info Platforms** **Test: Info Platforms**

	precision	recall	f1-score	support
Buruli Ulcer	0.882	0.862	0.872	130
Chagas disease (American trypanosomiasis)	0.868	0.750	0.805	88
Chromoblastomycosis	0.143	0.083	0.105	12
Dracunculiasis (guinea-worm disease)	0.562	0.290	0.383	31
Human African trypanosomiasis (HAT) - Sleeping sickness	0.556	0.283	0.375	53
Leishmaniasis	0.849	0.799	0.823	169
Leprosy	0.993	0.992	0.992	3495
Lymphatic filariasis (Elephantiasis)	0.899	0.863	0.880	226
Mycetoma	0.739	0.395	0.515	43
Noma	0.000	0.000	0.000	7
Onchocerciasis	0.866	0.815	0.840	119
Podoconiosis	0.854	0.761	0.805	46
Scabies	0.273	0.083	0.128	36
Schistosomiasis	0.934	0.876	0.904	226
Snakebite envenoming	0.922	0.855	0.887	69
Soil-transmitted helminths	0.926	0.849	0.886	192
Trachoma	0.901	0.820	0.858	122
Yaws	0.645	0.426	0.513	47
Zika virus	0.733	0.579	0.647	19
micro avg	0.956	0.917	0.936	5130
macro avg	0.713	0.599	0.643	5130
weighted avg	0.945	0.917	0.929	5130
samples avg	0.973	0.965	0.965	5130

Appendix C.6 - Disease multilabel classification

Model: BERT **Train: Info Platforms** **Test: Combined**

	precision	recall	f1-score	support
Buruli Ulcer	0.918	0.892	0.905	213
Chagas disease (American trypanosomiasis)	0.976	0.689	0.808	1987
Chromoblastomycosis	0.143	0.010	0.018	105
Dracunculiasis (guinea-worm disease)	0.562	0.090	0.155	100
Human African trypanosomiasis (HAT) - Sleeping sickness	0.789	0.080	0.145	939
Leishmaniasis	0.982	0.679	0.803	3887
Leprosy	0.493	0.992	0.659	3557
Lymphatic filariasis (Elephantiasis)	0.777	0.755	0.766	286
Mycetoma	0.778	0.087	0.157	241
Noma	0.000	0.000	0.000	60
Onchocerciasis	0.955	0.780	0.859	576
Podoconiosis	0.673	0.761	0.714	46
Scabies	0.273	0.007	0.014	421
Schistosomiasis	0.956	0.803	0.873	1585
Snakebite envenoming	0.993	0.629	0.770	1206
Soil-transmitted helminths	0.867	0.683	0.764	612
Trachoma	0.958	0.794	0.868	407
Yaws	0.667	0.264	0.378	91
Zika virus	0.995	0.688	0.813	1223
micro avg	0.751	0.694	0.721	17542
macro avg	0.724	0.509	0.551	17542
weighted avg	0.829	0.694	0.705	17542
samples avg	0.687	0.698	0.689	17542

Appendix C.7 - Disease multilabel classification

Model: BERT **Train: Combined** **Test: Info Platforms**

	precision	recall	f1-score	support
Buruli Ulcer	0.857	0.877	0.867	130
Chagas disease (American trypanosomiasis)	0.807	0.761	0.784	88
Chromoblastomycosis	0.400	0.500	0.444	12
Dracunculiasis (guinea-worm disease)	0.625	0.645	0.635	31
Human African trypanosomiasis (HAT) - Sleeping sickness	0.679	0.717	0.697	53
Leishmaniasis	0.811	0.811	0.811	169
Leprosy	0.992	0.993	0.993	3495
Lymphatic filariasis (Elephantiasis)	0.877	0.885	0.881	226
Mycetoma	0.703	0.605	0.650	43
Noma	1.000	0.857	0.923	7
Onchocerciasis	0.817	0.824	0.820	119
Podoconiosis	0.766	0.783	0.774	46
Scabies	0.500	0.444	0.471	36
Schistosomiasis	0.902	0.898	0.900	226
Snakebite envenoming	0.897	0.884	0.891	69
Soil-transmitted helminths	0.910	0.896	0.903	192
Trachoma	0.888	0.844	0.866	122
Yaws	0.633	0.660	0.646	47
Zika virus	0.688	0.579	0.629	19
micro avg	0.940	0.939	0.940	5130
macro avg	0.776	0.761	0.768	5130
weighted avg	0.940	0.939	0.939	5130
samples avg	0.982	0.981	0.977	5130

Appendix C.8 - Disease multilabel classification

Model: BERT **Train: Combined** **Test: Combined**

	precision	recall	f1-score	support
Buruli Ulcer	0.907	0.915	0.911	213
Chagas disease (American trypanosomiasis)	0.977	0.972	0.974	1987
Chromoblastomycosis	0.895	0.895	0.895	105
Dracunculiasis (guinea-worm disease)	0.871	0.810	0.839	100
Human African trypanosomiasis (HAT) - Sleeping sickness	0.962	0.949	0.955	939
Leishmaniasis	0.982	0.978	0.980	3887
Leprosy	0.990	0.991	0.991	3557
Lymphatic filariasis (Elephantiasis)	0.883	0.895	0.889	286
Mycetoma	0.952	0.896	0.923	241
Noma	0.982	0.933	0.957	60
Onchocerciasis	0.951	0.944	0.948	576
Podoconiosis	0.766	0.783	0.774	46
Scabies	0.940	0.924	0.932	421
Schistosomiasis	0.963	0.958	0.960	1585
Snakebite envenoming	0.986	0.974	0.980	1206
Soil-transmitted helminths	0.937	0.928	0.933	612
Trachoma	0.948	0.948	0.948	407
Yaws	0.780	0.780	0.780	91
Zika virus	0.989	0.984	0.986	1223
micro avg	0.971	0.966	0.968	17542
macro avg	0.930	0.919	0.924	17542
weighted avg	0.971	0.966	0.968	17542
samples avg	0.981	0.981	0.979	17542

Appendix C.9 - Disease multilabel classification

Model: Gemini 2.0 Flash Train: N/A Test: Info Platforms

	precision	recall	f1-score	support
Buruli Ulcer	0.829	0.823	0.826	130
Chagas disease (American trypanosomiasis)	0.684	0.761	0.720	88
Chromoblastomycosis	0.240	0.500	0.324	12
Dracunculiasis (guinea-worm disease)	0.516	0.516	0.516	31
Human African trypanosomiasis (HAT) - Sleeping sickness	0.571	0.604	0.587	53
Leishmaniasis	0.631	0.769	0.693	169
Leprosy	0.993	0.967	0.980	3495
Lymphatic filariasis (Elephantiasis)	0.824	0.805	0.814	226
Mycetoma	0.622	0.535	0.575	43
Noma	0.318	1.000	0.483	7
Onchocerciasis	0.702	0.672	0.687	119
Podoconiosis	0.650	0.848	0.736	46
Scabies	0.383	0.500	0.434	36
Schistosomiasis	0.827	0.823	0.825	226
Snakebite envenoming	0.857	0.870	0.863	69
Soil-transmitted helminths	0.797	0.797	0.797	192
Trachoma	0.797	0.803	0.800	122
Yaws	0.585	0.660	0.620	47
Zika virus	0.522	0.632	0.571	19
micro avg	0.901	0.902	0.902	5130
macro avg	0.650	0.731	0.676	5130
weighted avg	0.911	0.902	0.905	5130
samples avg	0.940	0.962	0.945	5130

Appendix C.10 - Disease multilabel classification

Model: Gemini 2.0 Flash Train: N/A Test: Combined

	precision	recall	f1-score	support
Buruli Ulcer	0.896	0.887	0.892	213
Chagas disease (American trypanosomiasis)	0.934	0.975	0.954	1987
Chromoblastomycosis	0.817	0.895	0.855	105
Dracunculiasis (guinea-worm disease)	0.821	0.780	0.800	100
Human African trypanosomiasis (HAT) - Sleeping sickness	0.918	0.930	0.924	939
Leishmaniasis	0.958	0.976	0.967	3887
Leprosy	0.990	0.965	0.978	3557
Lymphatic filariasis (Elephantiasis)	0.623	0.808	0.703	286
Mycetoma	0.894	0.664	0.762	241
Noma	0.783	0.900	0.837	60
Onchocerciasis	0.920	0.913	0.916	576
Podoconiosis	0.500	0.848	0.629	46
Scabies	0.924	0.898	0.911	421
Schistosomiasis	0.940	0.940	0.940	1585
Snakebite envenoming	0.990	0.900	0.943	1206
Soil-transmitted helminths	0.787	0.850	0.817	612
Trachoma	0.923	0.916	0.920	407
Yaws	0.755	0.813	0.783	91
Zika virus	0.990	0.975	0.982	1223
micro avg	0.939	0.942	0.940	17542
macro avg	0.861	0.886	0.869	17542
weighted avg	0.942	0.942	0.941	17542
samples avg	0.941	0.961	0.947	17542

Appendix C.11 - Disease multilabel classification

Model: SVM + BERT + Gemini prediction union **Train:Info Platforms** **Test:Info Platforms**

	precision	recall	f1-score	support
Buruli Ulcer	0.777	0.938	0.850	130
Chagas disease (American trypanosomiasis)	0.667	0.886	0.761	88
Chromoblastomycosis	0.241	0.583	0.341	12
Dracunculiasis (guinea-worm disease)	0.532	0.806	0.641	31
Human African trypanosomiasis (HAT) - Sleeping sickness	0.530	0.830	0.647	53
Leishmaniasis	0.631	0.953	0.759	169
Leprosy	0.984	0.997	0.991	3495
Lymphatic filariasis (Elephantiasis)	0.779	0.965	0.862	226
Mycetoma	0.613	0.884	0.724	43
Noma	0.318	1.000	0.483	7
Onchocerciasis	0.673	0.899	0.770	119
Podoconiosis	0.646	0.913	0.757	46
Scabies	0.410	0.694	0.515	36
Schistosomiasis	0.811	0.947	0.873	226
Snakebite envenoming	0.783	0.942	0.855	69
Soil-transmitted helminths	0.771	0.948	0.850	192
Trachoma	0.755	0.984	0.854	122
Yaws	0.579	0.936	0.715	47
Zika virus	0.520	0.684	0.591	19
micro avg	0.874	0.974	0.921	5130
macro avg	0.633	0.884	0.728	5130
weighted avg	0.896	0.974	0.930	5130
samples avg	0.959	0.990	0.967	5130

Appendix C.12 - Disease multilabel classification

Model: SVM + BERT + Gemini prediction union **Train: Combined** **Test:Info Platforms**

	precision	recall	f1-score	support
Buruli Ulcer	0.668	0.946	0.783	130
Chagas disease (American trypanosomiasis)	0.667	0.864	0.752	88
Chromoblastomycosis	0.276	0.667	0.390	12
Dracunculiasis (guinea-worm disease)	0.400	0.903	0.554	31
Human African trypanosomiasis (HAT) - Sleeping sickness	0.568	0.868	0.687	53
Leishmaniasis	0.624	0.923	0.745	169
Leprosy	0.981	0.997	0.989	3495
Lymphatic filariasis (Elephantiasis)	0.749	0.951	0.838	226
Mycetoma	0.565	0.907	0.696	43
Noma	0.318	1.000	0.483	7
Onchocerciasis	0.615	0.924	0.738	119
Podoconiosis	0.618	0.913	0.737	46
Scabies	0.338	0.722	0.460	36
Schistosomiasis	0.778	0.947	0.854	226
Snakebite envenoming	0.753	0.971	0.848	69
Soil-transmitted helminths	0.764	0.964	0.853	192
Trachoma	0.657	0.975	0.785	122
Yaws	0.449	0.936	0.607	47
Zika virus	0.483	0.737	0.583	19
micro avg	0.848	0.975	0.907	5130
macro avg	0.593	0.901	0.704	5130
weighted avg	0.881	0.975	0.920	5130
samples avg	0.954	0.991	0.964	5130

Appendix C.13 - Disease multilabel classification

Model: SVM + BERT + Gemini majority voting **Train: Info Platforms** **Test:Info Platforms**

	precision	recall	f1-score	support
Buruli Ulcer	0.896	0.862	0.878	130
Chagas disease (American trypanosomiasis)	0.852	0.784	0.817	88
Chromoblastomycosis	0.545	0.500	0.522	12
Dracunculiasis (guinea-worm disease)	0.760	0.613	0.679	31
Human African trypanosomiasis (HAT) - Sleeping sickness	0.727	0.604	0.660	53
Leishmaniasis	0.802	0.817	0.809	169
Leprosy	0.995	0.993	0.994	3495
Lymphatic filariasis (Elephantiasis)	0.922	0.885	0.903	226
Mycetoma	0.857	0.558	0.676	43
Noma	0.875	1.000	0.933	7
Onchocerciasis	0.911	0.773	0.836	119
Podoconiosis	0.844	0.826	0.835	46
Scabies	0.750	0.417	0.536	36
Schistosomiasis	0.943	0.881	0.911	226
Snakebite envenoming	0.954	0.899	0.925	69
Soil-transmitted helminths	0.933	0.865	0.897	192
Trachoma	0.937	0.852	0.893	122
Yaws	0.786	0.702	0.742	47
Zika virus	0.706	0.632	0.667	19
micro avg	0.962	0.935	0.948	5130
macro avg	0.842	0.761	0.795	5130
weighted avg	0.959	0.935	0.946	5130
samples avg	0.981	0.980	0.977	5130

Appendix C.14 - Disease multilabel classification

Model: SVM + BERT + Gemini majority voting **Train: Combined** **Test:Info Platforms**

	precision	recall	f1-score	support
Buruli Ulcer	0.844	0.877	0.860	130
Chagas disease (American trypanosomiasis)	0.769	0.795	0.782	88
Chromoblastomycosis	0.353	0.500	0.414	12
Dracunculiasis (guinea-worm disease)	0.571	0.645	0.606	31
Human African trypanosomiasis (HAT) - Sleeping sickness	0.692	0.679	0.686	53
Leishmaniasis	0.734	0.834	0.781	169
Leprosy	0.993	0.984	0.989	3495
Lymphatic filariasis (Elephantiasis)	0.893	0.889	0.891	226
Mycetoma	0.714	0.581	0.641	43
Noma	1.000	1.000	1.000	7
Onchocerciasis	0.789	0.815	0.802	119
Podoconiosis	0.822	0.804	0.813	46
Scabies	0.514	0.528	0.521	36
Schistosomiasis	0.906	0.898	0.902	226
Snakebite envenoming	0.923	0.870	0.896	69
Soil-transmitted helminths	0.938	0.865	0.900	192
Trachoma	0.837	0.885	0.861	122
Yaws	0.625	0.745	0.680	47
Zika virus	0.722	0.684	0.703	19
micro avg	0.936	0.935	0.936	5130
macro avg	0.771	0.783	0.775	5130
weighted avg	0.939	0.935	0.937	5130
samples avg	0.967	0.975	0.967	5130

Appendix C.15 - Disease multilabel classification

Model: SVM + BERT + Gemini prediction union **Train:Info Platforms** **Test: Combined**

	precision	recall	f1-score	support
Buruli Ulcer	0.843	0.958	0.897	213
Chagas disease (American trypanosomiasis)	0.926	0.981	0.953	1987
Chromoblastomycosis	0.798	0.905	0.848	105
Dracunculiasis (guinea-worm disease)	0.777	0.870	0.821	100
Human African trypanosomiasis (HAT) - Sleeping sickness	0.898	0.945	0.921	939
Leishmaniasis	0.953	0.985	0.969	3887
Leprosy	0.372	0.997	0.541	3557
Lymphatic filariasis (Elephantiasis)	0.612	0.948	0.743	286
Mycetoma	0.847	0.734	0.787	241
Noma	0.786	0.917	0.846	60
Onchocerciasis	0.893	0.960	0.926	576
Podoconiosis	0.506	0.913	0.651	46
Scabies	0.910	0.914	0.912	421
Schistosomiasis	0.930	0.960	0.945	1585
Snakebite envenoming	0.982	0.905	0.942	1206
Soil-transmitted helminths	0.768	0.923	0.838	612
Trachoma	0.894	0.973	0.932	407
Yaws	0.707	0.956	0.813	91
Zika virus	0.989	0.976	0.983	1223
micro avg	0.699	0.965	0.811	17542
macro avg	0.810	0.933	0.856	17542
weighted avg	0.807	0.965	0.856	17542
samples avg	0.785	0.970	0.845	17542

Appendix C.16 - Disease multilabel classification

Model: SVM + BERT + Gemini prediction union **Train: Combined** **Test: Combined**

	precision	recall	f1-score	support
Buruli Ulcer	0.759	0.962	0.849	213
Chagas disease (American trypanosomiasis)	0.928	0.983	0.955	1987
Chromoblastomycosis	0.800	0.914	0.853	105
Dracunculiasis (guinea-worm disease)	0.655	0.910	0.762	100
Human African trypanosomiasis (HAT) - Sleeping sickness	0.904	0.981	0.941	939
Leishmaniasis	0.950	0.988	0.969	3887
Leprosy	0.970	0.995	0.982	3557
Lymphatic filariasis (Elephantiasis)	0.608	0.958	0.744	286
Mycetoma	0.856	0.959	0.904	241
Noma	0.763	0.967	0.853	60
Onchocerciasis	0.866	0.976	0.918	576
Podoconiosis	0.488	0.913	0.636	46
Scabies	0.866	0.952	0.907	421
Schistosomiasis	0.920	0.976	0.947	1585
Snakebite envenoming	0.962	0.989	0.975	1206
Soil-transmitted helminths	0.778	0.969	0.863	612
Trachoma	0.838	0.990	0.908	407
Yaws	0.592	0.956	0.731	91
Zika virus	0.976	0.986	0.981	1223
micro avg	0.913	0.983	0.947	17542
macro avg	0.815	0.964	0.878	17542
weighted avg	0.920	0.983	0.949	17542
samples avg	0.963	0.988	0.970	17542

Appendix C.17 - Disease multilabel classification

Model: SVM + BERT + Gemini majority voting   Train: Info Platforms   Test:Combined

	precision	recall	f1-score	support
Buruli Ulcer	0.932	0.897	0.914	213
Chagas disease (American trypanosomiasis)	0.980	0.754	0.852	1987
Chromoblastomycosis	0.902	0.438	0.590	105
Dracunculiasis (guinea-worm disease)	0.891	0.490	0.632	100
ıuman African trypanosomiasis (HAT) - Sleeping sickness	0.946	0.445	0.605	939
Leishmaniasis	0.980	0.757	0.854	3887
Leprosy	0.525	0.993	0.686	3557
Lymphatic filariasis (Elephantiasis)	0.801	0.773	0.786	286
Mycetoma	0.933	0.402	0.562	241
Noma	0.976	0.683	0.804	60
Onchocerciasis	0.967	0.809	0.881	576
Podoconiosis	0.679	0.826	0.745	46
Scabies	0.975	0.466	0.630	421
Schistosomiasis	0.964	0.833	0.894	1585
Snakebite envenoming	0.996	0.700	0.822	1206
Soil-transmitted helminths	0.874	0.678	0.764	612
Trachoma	0.969	0.853	0.907	407
Yaws	0.865	0.703	0.776	91
Zika virus	0.995	0.877	0.932	1223
micro avg	0.794	0.787	0.790	17542
macro avg	0.903	0.704	0.770	17542
weighted avg	0.876	0.787	0.799	17542
samples avg	0.769	0.793	0.775	17542

Appendix C.18 - Disease multilabel classification

Model: SVM + BERT + Gemini majority voting   Train: Combined   Test:Combined

	precision	recall	f1-score	support
Buruli Ulcer	0.899	0.915	0.907	213
Chagas disease (American trypanosomiasis)	0.973	0.972	0.972	1987
Chromoblastomycosis	0.887	0.895	0.891	105
Dracunculiasis (guinea-worm disease)	0.844	0.810	0.827	100
Human African trypanosomiasis (HAT) - Sleeping sickness	0.958	0.956	0.957	939
Leishmaniasis	0.979	0.978	0.978	3887
Leprosy	0.992	0.982	0.987	3557
Lymphatic filariasis (Elephantiasis)	0.838	0.885	0.861	286
Mycetoma	0.941	0.867	0.903	241
Noma	1.000	0.950	0.974	60
Onchocerciasis	0.943	0.941	0.942	576
Podoconiosis	0.755	0.804	0.779	46
Scabies	0.953	0.914	0.933	421
Schistosomiasis	0.962	0.958	0.960	1585
Snakebite envenoming	0.995	0.965	0.980	1206
Soil-transmitted helminths	0.927	0.912	0.919	612
Trachoma	0.933	0.958	0.945	407
Yaws	0.778	0.846	0.811	91
Zika virus	0.995	0.982	0.988	1223
micro avg	0.969	0.962	0.966	17542
macro avg	0.924	0.921	0.922	17542
weighted avg	0.969	0.962	0.966	17542
samples avg	0.971	0.976	0.971	17542

Appendix C.19 - Disease multilabel classification full model summary:

Test set	Model	Train Set	Micro-F1	Macro-F1
Info Platforms	SVM	Info Platforms	**0.949**	**0.840**
		Combined	0.919	0.774
	BERT	Info Platforms	0.936	0.643
		Combined	0.940	0.768
	Gemini	N/A	0.902	0.676
	SVM + BERT + Gemini Union	Info Platforms	0.921	0.728
		Combined	0.907	0.704
	SVM + BERT + Gemini Majority Voting	Info Platforms	0.948	0.795
		Combined	0.936	0.775
Combined	SVM	Info Platforms	0.696	0.744
		Combined	0.954	0.902
	BERT	Info Platforms	0.721	0.551
		Combined	**0.968**	**0.924**
	Gemini	N/A	0.940	0.869
	SVM + BERT + Gemini Union	Info Platforms	0.811	0.856
		Combined	0.946	0.878
	SVM + BERT + Gemini Majority Voting	Info Platforms	0.790	0.770
		Combined	0.966	0.922

Appendix C.20 - Potential article count growth per disease for the Info Platforms using SVMs models on the PubMed 2025 Annual Baseline dataset.

Disease	# Total Identified	% Newly Predicted (...	% Matched (n)	# Growth Factor
Zika virus	13189	98.8% (13037)	1.2% (152)	86.77
Chagas disease (American trypanosomiasis)	14772	97.1% (14340)	2.9% (432)	34.19
Leishmaniasis	26660	96.5% (25728)	3.5% (932)	28.61
Snakebite envenoming	7724	94.9% (7330)	5.1% (394)	19.6
Scabies	5512	94.2% (5193)	5.8% (319)	17.28
Human African trypanosomiasis (HAT) - Sleeping sickness	4615	94.0% (4339)	6.0% (276)	16.72
Schistosomiasis	13483	92.1% (12420)	7.9% (1063)	12.68
Noma	287	88.5% (254)	11.5% (33)	8.7
Soil-transmitted helminths	5549	84.8% (4706)	15.2% (843)	6.58
Onchocerciasis	4832	83.8% (4049)	16.2% (783)	6.17
Trachoma	5444	83.3% (4536)	16.7% (908)	6.0
Mycetoma	1646	81.7% (1345)	18.3% (301)	5.47
Chromoblastomycosis	333	76.3% (254)	23.7% (79)	4.22
Dracunculiasis (guinea-worm disease)	1510	75.2% (1135)	24.8% (375)	4.03
Buruli Ulcer	2498	75.1% (1877)	24.9% (621)	4.02
Lymphatic filariasis (Elephantiasis)	3639	75.1% (2732)	24.9% (907)	4.01
Yaws	1659	72.3% (1199)	27.7% (460)	3.61
Leprosy	57160	60.4% (34512)	39.6% (22648)	2.52
Podoconiosis	186	26.3% (49)	73.7% (137)	1.36